DETECTION OF FALLEN ELECTRIC CABLESON ROAD AND STREETS IN INNER CITIES FOR AUTONOMOUS VEHICLES AND BIKES

UBRAN FALLEN CABLE DETECTION DATASET (UFCDD): A VISUAL BENCHMARK FOR SAMRT MOBILITY IN PAKISTAN

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CERTIFICATION

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DEDICATION

This work is dedicated to my family, especially my parents, wife, and my beloved kids for continuous support and prayers.

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Humza Nazeer

ABSTRACT*

Abstract

Robust perception systems are essential for autonomous vehicles (AVs) to operate safely in intricate urban settings. Even though modern perception frameworks perform well for typical road entities, they are nevertheless susceptible to High-Stakes, Low-Frequency (HSLF) risks, especially thin, irregular objects like fallen communication and power cables. These risks, which include electrical failure, vehicle damage, and possible loss of control, are particularly common in developing areas' unstructured urban contexts where infrastructure maintenance is irregular. Such risks are mostly ignored by current autonomous driving datasets and detection methods, which leads to poor generalization and high false-negative rates. This study proposes a specific deep learning-based detection system for falling cable dangers in order to close this crucial safety gap. First, a brand-new dataset called the Urban Fallen Cable Detection Dataset (UFCDD) is created. It consists of more than 2,500 meticulously annotated photos taken in various lighting and weather situations from actual urban settings in Pakistan. Second, the YOLOv5 architecture is used to create a specific Convolutional Neural Network (CNN) based object detection model that is tuned for real-time detection of thin, curvilinear structures in chaotic situations. In order to facilitate downstream navigation and path planning, the framework prioritizes accurate localization and categorizes cable conditions as either safe or unsafe. The suggested method obtains a mean Average Precision of 96% at mAP@50 and 76% at the stricter mAP@50–95, according to extensive experimental evaluation, demonstrating its efficacy in low-aspect-ratio danger detection and fine-grained localization. These findings confirm the viability of accurate thin-structure detection in safety-critical autonomous driving scenarios and show a notable improvement over generic perception pipelines. All things considered, this work offers a novel dataset, a verified real-time detection system, and empirical evidence supporting improved AV safety and scalability, especially for

implementation in developing nations with unstructured and infrastructure-challenged urban environments.

Chapter No. 01

INTRODUCTION

1.1 Background

1.1.1 The Advancement of Autonomous Mobility and Perception Systems

The creation of the autonomous vehicle (AV), which is currently regarded as a key element of contemporary transportation systems, is the result of a determined concentration over the past few years on improving technology to speed up tasks and enhance human life. Mobility, efficiency, and public safety are all expected to significantly improve with autonomous cars, which are categorized into six levels of automation (SAE Levels 0–5). Strong safety statistics support the need for this technology: according to World Health Organization reports, human error is a contributing factor in the great majority of traffic collisions, which cause over 1.35 million lives worldwide each year. By minimizing human factors such as fatigue, distraction, or impaired driving, AVs offer the potential to drastically reduce these catastrophic failures. Additionally, the use of AVs promotes environmental sustainability through improved traffic flow and lower emissions, and it increases inclusivity by giving mobility to populations that cannot drive [1].

An AV's perception system, which is responsible for the real-time recognition, categorization, and localization of objects in the surrounding environment, is essential to its operational integrity [2]. Sensor fusion, which integrates input from several modalities, is crucial to this intricate capability. Rich visual and semantic information about color, texture, and lane markers is provided via cameras. LiDAR helps in mapping and localization by providing high-fidelity three-dimensional (3D) point clouds that are essential for understanding depth and space. Strong distance and speed readings are provided by radar systems, which are very helpful in bad weather when optical sensors have trouble. Convolutional Neural Networks (CNNs) and CNN-Transformer hybrids, which automatically extract hierarchical features from raw data to identify things like cars, people, and traffic signs, are the most advanced deep learning frameworks that analyze these inputs [3].

Modern perception systems have made remarkable progress, especially in identifying typical, structured road elements. However, when these algorithms come across geometrically complicated, irregular, or unconventional hazards, their performance deteriorates [4]. The design efficiency of existing sensor fusion systems is a significant restriction; whereas sensors offer redundancy for large objects, such as radar compensating for camera failures in fog, they show limits versus small structures. For thin objects, LiDAR frequently produces sparse point cloud returns, while radar cross-sections are too small for accurate detection. In order to make up for the lack of trustworthy complementing depth data from other sensors, the perception framework is forced to rely almost exclusively on high-resolution monocular vision, requiring the accompanying detection algorithm to attain extraordinary precision and robustness against background clutter. The impression that systems must be resilient to unpredictable settings and infrastructure failures is therefore necessary for the effective deployment of AVs globally, prompting focused study into these neglected hazard areas.

1.1.2 **The High-Stakes, Low-Frequency Hazard of Fallen Cables**

One crucial class of High-Stakes, Low-Frequency (HSLF) risks is represented by fallen electrical and communication wires on roads. Failing to identify these objects can have serious consequences, such as serious mechanical damage to the vehicle's undercarriage, complete system failure from electrical shorts, and possibly loss of control that puts the public and occupants in danger [5]. The high risk of cable-related infrastructure failures is demonstrated by the fatality rates linked to power line incidents in other transportation domains, such as the 30% fatality rate reported by the Federal Aviation Administration for aircraft-power line collisions between 1997 and 2010.

This problem is made worse in unstructured urban settings, especially in developing nations where density is high and infrastructure repair is frequently erratic. Pakistan is expressly mentioned in the research material, which notes that such environments provide particular operational issues that are not usually addressed by global standards that are Western-focused [6]. Electrical lines breaking and falling into the roads is also directly caused by environmental variables, such as the increased frequency of extreme weather events like dust storms as a result of climate change [7]. AVs must exhibit dependable performance against these regionally specific infrastructure and environmental failures in order to obtain worldwide safety certification and scalable deployment.

1.2 Problem Statement

A fundamental and important safety concern that jeopardizes the dependability and worldwide scalability of autonomous vehicle technology is the lack of dependable, real-time detecting techniques for fallen electric cables in highly cluttered metropolitan environments.

Low-hanging and fallen electric cables are a serious risk since they are hazardous road hazards that can impede traffic, harm complex car parts, and put people and the vehicle's electrical systems at risk of electrocution [5]. The intrinsic intricacy of these things causes existing perception systems to fail; they are difficult for generalized vision apparatus to detect, especially in the complex and dynamic visual background of an inner metropolis [7]. These cables, which frequently cover just a few pixels in a camera image, are characterized geometrically by a dangerously low aspect ratio. This leads to visual uncertainty, which makes them mistaken for benign linear lines, background noise, or shadows. This results in significant false negative rates, when the hazard is completely overlooked.

A serious lack of training resources exacerbates this technological flaw. Thin, irregular dangers like falling electric cables are entirely ignored by standard autonomous vehicle databases, which heavily favor common traffic features like pedestrians, cars, and road signs [8]. As a result, it is unable to properly train and evaluate specific deep learning models that can recognize the fine-grained, curvilinear properties of these objects. A major safety gap is created by the absence of specialized data and customized detection models for thin, irregular dangers.

To successfully identify these risks and guarantee safe autonomous vehicle navigation in complex, unstructured urban areas, the goal is to create a strong, deep learning-based detection framework that is backed by a unique, specialized dataset called the Urban Fallen Cable Detection Dataset (UFCDD).

1.3 Research Questions

By directing the methodical creation of the necessary data and detection framework, the following research questions are intended to address the issues mentioned in the Problem Statement.

RQ1: Data Generation and Fidelity: In unstructured urban settings, how can a dedicated, high-fidelity visual dataset (UFCDD) be methodically gathered and annotated to efficiently capture the various geometric and photometric characteristics of fallen and low-hanging electric cables across a range of environmental (clear, overcast) and lighting conditions (dawn, dusk, day)?

In contrast to earlier benchmarks, this question shows the fundamental necessity of developing a dataset that adequately depicts the real-world complexity and unpredictability of cable hazards in the target domain [1].

RQ2: Model Architecture and Optimization: Can the customized Urban Fallen Cable identification Dataset (UFCDD) be used to robustly train a Convolutional Neural Network (CNN)-based object identification framework, specifically designed using the YOLOv5 architecture, to accomplish real-time, high-accuracy detection of thin, curvilinear cable hazards?

This inquiry investigates whether CNNs, which are renowned for their capacity to extract features, can effectively address the technical challenge presented by thin structures. Because AV systems need low latency for safety-critical actions, the choice of YOLOv5, a cutting-edge single-stage detector, is implicitly crucial [2]. The idea behind the architecture is that YOLOv5 provides the speed and accuracy balance required to convert detection into practical, real-time safety measures.

RQ3: Performance Validation and Localization Precision: What degree of real-time performance (measured by metrics like mAP@50 and the stricter mAP@50-95) can the specialized YOLOv5 framework attain in precisely identifying and categorizing fallen cable hazards, proving a noticeable improvement in robustness against the geometric difficulties of thin structures?

The emphasis here is on thorough empirical validation. Since thin objects are difficult to locate precisely, it is crucial to evaluate the model using strict metrics such as mean Average Precision at Intersection over Union thresholds from 50% to 95% (mAP@50-95) in order to confirm not only detection but also precise boundary definition, which is required for efficient path planning around the hazard. The model's capacity to resolve the geometric ambiguity present in thin, low-pixel-count objects is confirmed by the localization accuracy.

RQ4: Impact on Safety and Scalability:

What are the consequences for scalable smart mobility solutions in poor nations where these risks are common, and how can the suggested detection framework, made possible by the UFCDD, improve autonomous vehicle safety and dependability in complicated urban environments?

By connecting the technical answer to societal advantages, this question tackles the research's practical application and wider relevance, particularly in contexts that are internationally representative of diverse infrastructural quality [6].

1.4 Research Contributions

The study makes several specialized additions to state-of-the-art in autonomous vehicle perception and safety, especially regarding unusual risks in unstructured situations.

1.4.1 Creation and Dissemination of a Novel Dataset

The creation and compilation of the Urban Fallen Cable Detection Dataset (UFCDD) is the main contribution. This dataset, which is mostly collected from Mirpur, AJK, Pakistan, is a unique collection of more than 2,500 carefully annotated photos that specifically show fallen and low-hanging electric and communication lines in urban areas. This collection directly addresses a crucial gap in current datasets, which have historically ignored these cable-induced risks in favor of concentrating on cars, pedestrians, and traffic signs [6]. To ensure that the final models are resilient to changing lighting and weather, the UFCDD is designed to incorporate photos taken in a variety of settings, including early morning, afternoon, and evening. This helps to mitigate frequent performance limitations in AV systems. The UFCDD offers the geographical and infrastructure diversity required for creating AV systems that can function securely on a worldwide scale by concentrating an area outside of the usual Western-centric AV testing domain.

1.4.2 Implementation of a Specialized High-Performance Detection Framework

A key addition is a specific deep learning object detection framework that makes use of the YOLOv5 architecture's accuracy and efficiency. To identify the space-based fine-grained features (edges, curves, and shapes) necessary for identifying thin, elongated objects in complex urban clutter, this framework relies on Convolutional Neural Networks' (CNNs) ability to automatically learn and extract a hierarchy of feature representations from raw image data. As a cutting-edge single-stage detector, YOLOv5 strikes a compromise between computational speed and detection

accuracy, which is essential for the low-latency inference needs of safety-critical autonomous driving [1, 7, 8, 9, 10]. The model is taught to distinguish between "Safe" (intact) and "Unsafe" (fallen or hazardous) cable circumstances, converting intricate visual data into straightforward inputs for navigational decision-making.

1.4.3 Quantitative Validation of Robust Thin-Structure Detection

The specific CNN-based YOLOv5 model's performance was effectively confirmed by the study, offering measurable proof of its effectiveness in identifying thin, irregular dangers. The model's accuracy was 76% at mAP@50-95 and 96% at mAP@50. Strong classification ability is demonstrated by a high mAP@50 score, which consistently separates dangerous wires from the background. Moreover, the strong 76% accuracy at the mAP@50-95 threshold is very noteworthy. The outcome demonstrates that the framework surmounted the fundamental geometric barrier presented by low-pixel-count objects, since this metric necessitates a very accurate border determination (high Intersection over Union scores). This effective validation contributes to a dependable detection system that improves urban traffic safety by demonstrating the customized model's capacity to recognize thin, uneven cable structures even in challenging urban environments [13].

Table 1.1: Key Contributions and Performance Benchmarks

Contribution Element	**Description/Metric**	**Significance**
Dataset Name	Urban Fallen Cable Detection Dataset (UFCDD)	Addresses overlooked High-Stakes, Low-Frequency (HSLF) hazards [14].
Dataset Size/Scope	2,500+ images from Mirpur, AJK, Pakistan	Provides critical geographical and infrastructural diversity for global AV deployment [15].

Detection Framework	CNN-based YOLOv5 Model	Ensures real-time, low-latency performance required for safety-critical AV tasks [16].
Mean Average Precision (mAP@50)	96%	High accuracy for primary detection threshold, indicating strong capability in distinguishing hazardous objects [17].
Mean Average Precision (mAP@50-95)	76%	Robust performance across stricter localization tolerances, confirming precision on thin, complex structures [18].

1.5. Research Objectives

A set of specific goals intended to methodically close the identified safety gap through data collection and model application serve as the research's compass.

1. **Hazard Image Collection:** To compile an extensive collection of photos showing downed power and communication lines from different Pakistani cities, making sure the information accurately represents the diversity of real-world infrastructure. This collection's scope is defined to encompass several regions of Pakistan (Mirpur) to account for differences in infrastructure and geography.
2. **Environmental Diversity Acquisition:** To collect information on a wide range of natural lighting and weather situations, including clear daylight, cloudy or overcast circumstances, and particular time periods like dawn, late morning, afternoon, and evening (dusk). This guarantees that the visual fluctuations typical of dynamic situations are used to train the model.
3. **Manual Annotation and Ground Truth Generation:** In order to produce trustworthy ground truth, the gathered imagery must be meticulously manually tagged, with a focus on

delineating the spatial boundaries of fallen cables. The supervised deep learning model's training depends on this procedure. The public

4. **Dataset Construction:** To organize, verify, and develop the Urban Fallen Cable Detection Dataset (UFCDD) as a publicly available resource to aid in the testing and training of hazard detection systems for autonomous cars and more general robotics applications, such as drone navigation and surveillance systems.
5. **Model Development and Validation:** To train a CNN-based object detection model (YOLOv5) on the UFCDD and quantitatively assess its real-time performance using well-known metrics like mAP@50, mAP@50-95, precision, recall, and F1-score, confirming its appropriateness for low-latency, safety-critical AV operation [11, 12, 13].

1.6. Research Gap

This specific research is justified by filling in a number of important gaps that together restrict the safety and dependability of autonomous cars, especially in different and unstructured metropolitan areas across the globe.

1.6.1 Inherent Geometric and Photometric Challenges in Detection

The special physics and geometry of falling cables pose a fundamental barrier to traditional computer vision systems, which is the root of the technical gap. Because of their extremely low aspect ratio, which frequently shows in camera images as sparse lines of just one to three pixels, cables are either functionally invisible or confusing to generalized object detectors. The precision required to properly encompass or categorize these small, sub-scale objects is lacking in traditional bounding box frameworks, which are designed for larger targets like cars. Additionally, there is a lot of photometric ambiguity in the visual scene. Fallen cables often blend easily with background noise, shadows, and harmless linear characteristics and show little distinction against complicated road surfaces (concrete, asphalt, dirt). Standard perception pipelines are not designed to handle this challenge, which necessitates the specific extraction of fine-grained information like faint edges and curves [18].

1.6.2 Deficiencies in Standard Autonomous Vehicle Datasets

The widespread lack of data in recognized benchmarks is one of the main obstacles to creating a reliable detection system. For this endeavor, standard AV datasets—including important public resources like KITTI, NuScenes, and BDD100K—are inadequate. The majority of these benchmarks were gathered in highly developed metropolitan or highway environments, which naturally reduced the likelihood of unforeseen infrastructure failures like utility outages or extensive debris. Two major shortcomings in the available data define the research gap:

1. **Annotation Scarcity:** The proper annotation and particular taxonomy required for thin, unconventional threats are absent from these widely used datasets. The creation of machine learning models that can generalize dynamic, highly cluttered situations is hindered by this lack of communication.
2. **Misaligned Debris Focus:** Although there are some specialized datasets (like HazardNet) for generalized road debris, they typically use height thresholds that specifically exclude the critically low-lying geometry of fallen electrical lines and concentrate on larger, non-curvilinear obstacles like tires or construction materials.

Models are forced to rely on characteristics learnt from irrelevant or benign contexts due to a lack of specialized training data, which leads to poor detection performance for HSLF cable risks. By selecting imagery unique to high-risk, unstructured urban road situations, the UFCDD immediately fills this data gap.

1.6.3 The Domain Transfer and Infrastructural Gap

Successful earlier research and the present need for ground-based autonomous vehicles have a significant operational gap. The use of unmanned aerial vehicles (UAVs) for aerial inspection of overhead power lines and pylons was a major focus of earlier line detection research [10]. These approaches used methods such as feature grouping, Kalman filtering, and Radon transform, which worked well in controlled settings with predictable backdrops [11]. Nevertheless, there are significant obstacles to the transfer of these airborne methods to ground AVs. Structured, straight-line geometries and great contrast (wire against a clear sky) are advantageous for aerial systems. Instead, ground AVs have to deal with a lot of background clutter, complicated road textures,

rainfall-induced visual interference, and the extremely complicated, curved, and non-rigid pathways of fallen cables. These topological and environmental differences require that detection algorithms be fundamentally adjusted. Additionally, the emphasis on unstructured urban settings in places like Pakistan draws attention to a basic infrastructure and geopolitical gap in the development of AV safety. The frequency of fallen cables in these locations is a foreseeable danger associated with elements like infrastructure density, sporadic maintenance, and environmental stress rather than a random anomaly [6]. The need for a region-specific dataset such as the UFCDD shows that global AV maturity (SAE Level 5) requires modular perception stacks that can reliably adapt to a variety of real-world infrastructure quality and regional hazards, rather than relying solely on centralized, generalized models. Scalable smart mobility solutions in developing nations are made possible by this research, which lays the groundwork for identifying and responding to these specific geographical shortcomings.

Chapter No. 02: Relevant Research in This Field

Chapter No. 3: Give a thorough description of the materials and experimental setups.

Chapter 4: Outcomes of research and recommended configurations.

Chapter No. 5: How to Expand This Research and Potential Future Directions for Research Extension.

Chapter No. 02

RELATED WORK

2.1. Foundations of Autonomous Vehicle Perception and System Architecture

In order to achieve completely autonomous mobility (SAE Levels 4 and 5), highly dependable perception systems that can achieve comprehensive environmental knowledge under all operational scenarios must be developed.[1] The perception system supplies the crucial, safety-critical data input for the ensuing decision-making processes in autonomous vehicles (AVs), which function on a linked stack of modules that includes perception, localization, planning, and control.[2]

2.1.1. Perception Requirements and Sensor Modalities

In order to overcome the inherent limits of any one type of sensor, robust perception systems in contemporary AVs rely on a multi-modal sensor suite.[3]

The Visual Modality (Cameras) is essential for understanding complicated scenes since it provides rich semantic information, color, texture, and lane markers.[1] To recognize objects in dynamic urban surroundings, camera data is fed into advanced deep learning architectures like You Only Look Once (YOLO) [42] and Faster R-CNN [13].[1] Despite being affordable and easy to use, vision systems are very vulnerable to performance degradation in difficult circumstances, such as low light, strong glare, fog, or rain, which jeopardizes image quality and system dependability.[7]

Visual data is supplemented by ranging modalities, mainly Light Detection and Ranging (LiDAR) and Radar. High-fidelity three-dimensional (3D) point clouds are produced by LiDAR, offering exact spatial, geometric, and distance information necessary for accurate localization and obstacle avoidance.[1] In contrast, radar performs exceptionally well in measuring speed and distance, providing greater resilience in unfavorable weather circumstances (rain, fog, dust) where optical sensors frequently malfunction [4].

2.1.2 Sensor Limitations in Detecting Thin Structures

When these conventional sensor modalities come into contact with thin, geometrically complicated materials, such fallen electric lines, a significant problem occurs. Range-based sensors are unreliable for fine wire detection due to their inherent characteristics.

A cable's narrow profile in relation to its length, or its critically low aspect ratio, causes a sparse point cloud return for LiDAR.[5] LiDAR processing techniques struggle to accurately detect the obstruction due to this lack of geometric information.[5] Additionally, radar's main drawback is its intrinsic poor resolution, which makes it unsuitable for this particular hazard assessment task because it doesn't provide enough detail to identify small, static, or fine impediments like falling cables.[5]

The vision-based system is disproportionately burdened by ranging sensors' incapacity to accurately detect these dangers.[5] As a result, in situations where geometric sensors are unable to extract the fine semantic properties (edges, shading, texture) of the cable structure, successful identification must heavily rely on high-resolution visual processing systems and specialized deep learning models.[5] The architectural limitations covered in following sections are driven by this required research focus.

2.2 The Critical Challenge: High-Stakes, Low-Frequency (HSLF) Obstacles

Even though the most advanced AV perception systems available today are very good at recognizing common, structured traffic elements, cars, pedestrians, and traffic signs, they are severely limited when it comes to uncommon, geometrically complex, or unusual road hazards, which are collectively referred to as road debris or infrastructure failures.[1]

2.2.1 Defining Unconventional Road Hazards and Infrastructure Failures

One important class of High-Stakes, Low-Frequency (HSLF) dangers is represented by fallen power and telecom cables on metropolitan roads.[1] Running over such an impediment could result in catastrophic consequences, such as serious mechanical damage to the vehicle's undercarriage, system failure from electrical shorts, and loss of control that puts both the public

and the occupants of the vehicle in danger.[1] Organizations like the American Automobile Association (AAA) and the Federal Aviation Administration (FAA) have highlighted the seriousness of collisions and crashes caused by power lines and road debris, confirming the need for reliable detection systems.[17]

In unstructured urban settings, such those found in poor nations like Mirpur, Pakistan, this safety issue is more severe.[25] The risk of electrical cables falling onto the road is increased in these environments due to high infrastructure density, inconsistent maintenance, and heightened susceptibility to environmental factors, particularly the increased frequency of severe weather events like dust storms that cause cable breakage.[25] Systems must be able to identify and respond to these region-specific infrastructural shortcomings in order for autonomous mobility to achieve global, scalable deployment.[25]

2.2.2 The Geometric Ambiguity Problem

Unlike conventional, huge obstructions, fallen power cables pose special geometric and photometric issues. Geometrically, they are characterized by a very low aspect ratio, which means that they are incredibly thin in relation to their length. In high-resolution camera photographs, they frequently span only a small number of pixels (occasionally 1 to 3 pixels), especially when viewed from a distance.[43]

Significant visual uncertainty and confusion are produced by this low pixel count. Cast shadows, rich road textures, unnecessary background noise, and benign linear features like painted markers or road fissures can all readily be mistaken for thin things.[45] Depending on the time of day or the weather, the cable's visual appearance (contrast, texture, color) in relation to the road surface varies significantly, creating low-contrast scenarios that confuse generalist detectors.[45]

This geometric complexity implies that when models are used in a high-clutter, unstructured environment, they experience an abrupt performance degradation known as domain shift. This is because models trained on structured datasets implicitly learn to ignore high levels of visual clutter.[45] After training on clean data, the model shows catastrophic generalization failure, completely ignoring the uncommon but crucial HSLF hazard [45].

Table 2.1 Thin-Object, Boundary-Aware, Powerline & Infrastructure Detection Research

Ref No.	Area / Focus	Key Findings	Relevance to Fallen Cable Detection
[9]	Aerial Power Line Extraction	Edge + region-based extraction	**Foundational thin-line detection** basis
[18]	Automated Power Line Inspection (Review)	**DL approaches** for line inspection	Techniques **transferable** to fallen cables
[36]	Power-line Segmentation	Semantic segmentation of overhead lines	**Directly adaptable** for fallen cable segmentation
[44]	Synthetic Wire Detection	**Synthetic → real** training improves robustness	Useful for **rare cable-fall training data**
[46]	Wire Detection + Depth	Cable and **depth estimation**	Useful for **locating** fallen wires on road
[51]	Small-object Sensitive Segmentation	**Attention** boosts thin-object sensitivity	**Critical** for long thin cable structures

[50]	Boundary-aware Loss Functions	Better **segmentation edges**	Helps separate cable pixels cleanly
[58]	**Thin-structure Obstacle Detection**	**Fast detection** of thin rods/wires	**Closest match** to fallen cable detection problem
[59]	Powerline & Pylon Detection	**Robust cable detection** from aerial views	**Cable shape priors** applicable to street-level detection

2.3 Evolution of Hazard Detection Methodologies

The need to accurately identify intricate, thin structures on the ground has spurred technological advancement, resulting in a major change from traditional computer vision to modern deep learning methods.

2.3.1 Classical Computer Vision Approaches

The inspection and identification of overhead power wires and pylons using imagery taken by unmanned aerial vehicles (UAVs) was the main focus of early utility infrastructure research.[1] These traditional approaches mostly depended on manually created, mathematically complex feature extraction. The Hough and Radon transforms, which were used to identify and recreate straight line segments based on predicted geometric features, were important methods.[48] These reconstructed line structures were then tracked across successive frames using methods like Kalman Filtering, which helped with inspection.[1]

However, when used in the ground-based domain of autonomous cars, these conventional approaches proven to be extremely brittle and unreliable. They were extremely sensitive to shadows, urban road texture variations, and background interference.[1] Most importantly, they were unable to predict the intricate, curving, and non-rigid routes that are typical of fallen cables

on the ground—a topology that is very different from the taut, structured straightness of overhead power lines.[1] The adoption of a more robust methodological paradigm was required due to this inability to generalize in complex ground clutter.[1]

2.3.2 The Paradigm Shift to Deep Learning

Deep learning (DL) techniques have become increasingly popular due to the inherent limitations of classical methodologies.[54] By automatically learning a hierarchy of complex patterns and nuanced features from raw picture data, Convolutional Neural Networks (CNNs) laid the groundwork for reliable and efficient identification.[1] This made it possible for systems to manage complicated background clutter and the visual variance brought on by shifting lighting and weather conditions, and it also removed the laborious process of manually constructing elements.[1]

The difficulty of domain transfer from aerial inspection to ground hazard detection is a major technical barrier in thin structure detection.[1] While ground-based AV systems must identify a dark cable against textured road surfaces that are frequently hidden by shadows or wetness, aerial detection benefits from high-contrast video (wire against the sky or clear background) [22].[1] Because of this intricacy, CNNs are uniquely able to learn a specialized feature set that focuses on ground texture and shadow distinction.

2.4 Deep Learning Architectures for Real-Time Object Detection in AVs

The safety-critical application of autonomous driving necessitates object detection systems with low latency (real-time inference) on edge computing hardware with limited resources.[1] High-efficiency architectural solutions are chosen due to this demand.

2.4.1 Taxonomy of Modern Detectors and the YOLO Framework

Two-stage (like Faster R-CNN) and single-stage (like YOLO, SSD) detectors are the two main categories of contemporary object identification algorithms.[1] Because they simplify the detection pipeline and achieve the fast inference speed (low latency) necessary for real-time safety maneuvers, single-stage detectors—especially those in the YOLO family (You Only Look Once)—are favored in AV perception over two-stage systems' higher latency.[7]

The YOLOv5 architecture, which was chosen for the suggested framework, is a cutting-edge model designed to strike a balance between speed and precision.[1] It consists of a detecting head, a neck network for multi-scale feature integration, and a sturdy backbone for feature extraction.[1] Compared to models tuned for single input resolutions, its intrinsic capacity to concurrently evaluate input images at several resolutions offers a structural advantage for recognizing small objects.[1]

2.4.2 Hybrid Models and Lightweight Backbones for Edge Computing

Specialized hybrid architectural frameworks are frequently needed due to the inherent tension between the need for low computing load (required for real-time deployment) and high precision (required for extracting minute cable details).[1]

Convolutional networks have been modernized with high-accuracy detection models like ConvNeXt, which achieve cutting-edge performance on par with Vision Transformers [2]. ConvNeXt is a great option for the high-precision refinement layer needed for thin object identification because of its resilience in extracting intricate feature maps.[59] On the other hand, models such as MobileNetV4 are lightweight architectures designed for low-latency inference and computational efficiency on edge devices [5]. A Universal Inverted Bottleneck (UIB) module significantly lowers the computational load while increasing speed [7].

In order to achieve the maximum detection precision needed for thin cable identification while preserving the lowest latency required for safety-critical operation, the architectural strategy combines the advantages of ConvNeXt (for feature robustness) and MobileNetV4 (for computational efficiency) [9].

Multi-Scale Feature Fusion is used for additional optimization. When fine features and spatial information are lost during the deep network down sampling process, Small Object Detection (SOD) suffers.[4] In order to mitigate feature distribution discrepancies between resolutions, architectures that use Feature Pyramid Networks (FPNs) within the neck network efficiently integrate low-level fine edge features (for precision) with high-level semantic information (for context).[61] In order to maintain the modest visual properties of a thin cable throughout the detection pipeline and improve localization accuracy, this feature aggregation approach is essential [4].

For detecting continuous abnormalities like long power lines, sophisticated hybrid models that combine CNNs for local feature detection and Transformers (using self-attention for global contextual comprehension) have also shown great success [55].

Table 2.2: Detection Models & Algorithms Relevant to Fallen Cable Detection

Ref No.	**Model / Method**	**Contribution / Insight**	**Relevance to Fallen Cable Detection**
[20]	**HOG + SVM** (Dalal & Triggs, 2005)	Classical gradient-based detection	Useful **baseline** for thin-object edges and cable contours
[21]	**Focal Loss** (Lin et al., 2017)	Handles imbalance and **small objects**	Fallen cables are **rare**, thin, elongated objects
[27]	Keypoint UAV Detection + **Rotated DIoU**	Detects rotated or elongated objects	Captures cable-like **long shapes** on roads
[28]	**LI-YOLO** (Low-light detection)	Strong detection in low illumination	Useful for **night-time** cable detection
[29]	Deep Learning Object Detection Survey	Overview of classical and modern detectors	Helps identify strong **baselines** for cable detection

[30]	**SSD**	Lightweight object detector	Suitable for **bike-mounted hardware**
[31]	**MobileNetV2**	Efficient convolutional backbone	**Real-time detection** on edge devices
[36]	Power-line Segmentation (Damodaran)	Segmentation of overhead lines	Applicable to fallen cable **geometry**
[44]	**Dilated CNN for Wire Detection**	Wire detection using **synthetic data**	Highly relevant for **rare-cable-event synthesis**
[51]	Small-object Sensitive Segmentation	**Across-feature-map attention**	Helps detect **thin cable-like structures**
[58]	**Thin-structure Obstacle Detection**	Detects thin rods/wires	Almost **directly applicable** to fallen wires
[59]	Cable & Pylon Detection	Vision-based cable detection (aircraft)	**Cable shape priors** transferable to road scenarios

[50]	**Boundary-aware Loss**	Improves fine object boundary segmentation	Useful for segmenting **cable edges** on road
[49]	**DL vs Traditional CV**	Comparative analysis	Helps **model-selection** for cable detection
[56]	**ConvNext + YOLOv5**	Strong small-object detection performance	Good for **thin-object detection**
[54]	**Loss Function Review**	Comprehensive loss analysis	Helps pick best loss for **thin-structure segmentation**

2.5 Specialized Techniques for Thin and Curvilinear Structure Perception

Because fallen cables are fundamentally geometric, detection techniques must put pixel-level boundary accuracy ahead of conventional bounding box localization.

2.5.1 Representation and Boundary Accuracy

Conventional bounding box detection frameworks, including Single Shot Detectors (SSDs), are frequently inadequate for correctly detecting objects that span only a few pixels since they are naturally designed for larger targets [23]. The bounding box result is insufficiently accurate in enclosing or classifying thin structures [24].

As a result, the pixel-wise categorization method known as **Semantic Segmentation** is considered the best approach for determining the exact physical extent of the risk [25]. Semantic segmentation accurately defines the contour of the item by giving each pixel in the image a categorical label. The vehicle can precisely perform an avoidance maneuver thanks to this pixel-

level granularity, which is essential for AV path planning [26]. In this field, basic models include architectures such as Fully Convolutional Networks (FCNs) and U-Net [27].

Instead of trying to segment each individual pixel, a more specialized method called **Key-Point Encoding** represents the curvilinear wire as a collection of uniformly distributed key points [1, (Liu et al., 2020)]. By streamlining the detection process, this simplification enables the network to precisely reconstruct the object curve using implicit functions, which are frequently used in frameworks intended for posture estimation and thin object recognition.

2.5.2 Contextual and Depth Requirements for Safety

The detection task for ground AVs is conceptually a hybrid task: the speed of object detection (YOLO) is needed for real-time responsiveness, but the safety of avoidance depends upon the pixel-level precision achieved through segmentation training principles.

Furthermore, given the visual ambiguity between a thin cable and a low-lying shadow or road crack [46], two-dimensional (2D) detection alone carries an intrinsic false positive/negative risk. To ensure the object is a true physical hazard, robust detection necessitates validation in 3D space [46]. Research into Joint Detection and Depth Estimation (e.g., UCorr) is crucial for ground AVs, as 3D information is required to distinguish a hazardous cable lying on the road from a non-hazardous linear marking [59]. Ultimately, achieving high-confidence safety often requires the integration of non-visual confirmation, such as monocular depth estimates or potentially specialized sensor inputs (e.g., magnetic field sensing for live wires), to increase redundancy and mitigate visual uncertainty [60].

2.6 Addressing Data Scarcity and Training Difficulties

Beyond building design, the difficulties in identifying uncommon, thin risks include basic problems with data quantity and quality, particularly with regard to extreme class imbalance (EVI).

2.6.1 The Problem of Extreme Class Imbalance (EVI)

At the pixel level, thin cable detection is essentially an EVI problem. The little number of pixels that constitute the thin wire (the positive class) is dwarfed by the vast majority of pixels that belong to the background (the negative class) in any given metropolitan image [46].

The network is biased toward predicting the dominant background class during training with traditional loss functions (such as Cross-Entropy) due to the large number of background samples [66]. This bias has serious consequences, including poor delineation of the few positive pixels that mark the cable boundary and high rates of false negatives, which cause the key hazard to be ignored [66].

2.6.2 Specialized Loss Functions for Robust Learning

Conventional loss functions must be supplemented or replaced with specialized alternatives that prioritize the few crucial cable pixels in order to lessen the systemic failure brought on by EVI [66].

The contribution of easily categorized negative examples (background) is greatly reduced by Focal Loss (FL), which dynamically re-weights the standard cross-entropy loss [48]. This method significantly increases accuracy in cluttered, unbalanced images by forcing the model's training to concentrate on the uncommon, hard negative examples, which are frequently found close to the object border [48].

For segmentation tasks involving extreme imbalance, Dice Loss and similar region-based losses (such as Tversky Loss and Generalized Dice Loss) are quite successful [66]. The goal of these losses is to maximize the spatial overlap (also known as the Intersection over Union, or IoU) between the ground truth and forecast regions [66]. Furthermore, specific boundary-aware loss functions are frequently required to compel the network to prioritize prediction confidence, particularly near object contours, since the misclassification of boundary pixels seriously impairs the perceived quality and safety utility of thin structure detection [66]. By applying a severe penalty for errors involving the minority (cable) class, the selection of these particular loss functions serves as an algorithmic safety guarantee and mitigates the statistical bias of the network.

2.6.3 Synthetic Data Generation (SDG)

The use of Synthetic Data Generation (SDG) techniques is required due to the practical difficulties in gathering adequate amounts of real-world imagery for HSLF events [26]. SDG is a viable option for unusual object classes, utilizing generative models such as Diffusion Models and Generative Adversarial Networks (GANs) or simulation-based rendering [15].

It has been demonstrated that pre-training models on artificially produced data with significant variation in pose, illumination, and occlusions, followed by fine-tuning on a small sample of real data, improves model resilience and speeds up development [66]. Crucially, the real-world dataset (the Urban Fallen Cable Detection Dataset, UFCDD) offers the environmental factors and realistic visual textures required for the target unstructured urban area (e.g., Mirpur, AJK, Pakistan) [66]. In order to ensure that SDG can provide variability without compromising domain realism and to enable efficient transfer learning, this real-world anchor is crucial for preserving integrity during synthetic scaling [15].

2.7 Analysis of Data Deficiencies and Domain Shift

Fundamental flaws in the training and validation datasets are directly responsible for the generalized AV perception systems' consistent inability to accurately identify falling cables.

2.7.1 Critical Review of Standard AV Benchmarks

KITTI, NuScenes, and BDD100K are globally known AV benchmark datasets that cover activities related to organized traffic environments (e.g., automobiles, pedestrians, lane lines) in highly established urban or highway settings [35].

In terms of thin dangers, these records show a serious systemic deficit [35]. Extremely thin objects that generate sparse returns are frequently systematically filtered out of the ground truth labeling process due to the strict quality assurance requirements during data curation, which frequently require a minimum number of confirming sensor returns (e.g., LiDAR points) to validate an annotation [44]. A widespread safety blind spot for thin, low-profile impediments is unintentionally created by this deliberate selection for traditional, well-defined targets, guaranteeing that models trained solely on this conventional data are specifically educated to avoid these HSLF threats [35].

2.7.2 Existing Datasets for Debris and Unstructured Ground

The particular risk of falling cables is not covered by alternative databases, even when they address other types of environmental complexity.

Unstructured Ground Datasets, like Kamino and the Robot Unstructured Ground Driving (RUGD) dataset, focus on scene semantics and general terrain drivability while addressing the difficulties of off-road or unpaved navigation [50]. These tools are useful for addressing excessive environmental clutter, but they do not provide the detailed, pixel-level annotation and particular taxonomy needed for infrastructure failures such as electric lines [50].

Larger, vehicle-damaging debris, such tires, bricks, or building materials, is given priority in road debris datasets (like HazardNet) [24]. The low-lying, curved infrastructural components that are essential to this study are essentially excluded by the hazard definition in these datasets, which frequently includes a minimum object height requirement (e.g., more than 15 cm) [24].

This investigation validates a distinct and crucial gap in the available public resources: no dataset currently in use offers both the specialized, required annotation taxonomy for fallen electric and communication cables and the necessary visual complexity (unstructured urban areas with thick clutter).

Table 2.3: Datasets Useful for Road Scene Analysis and Cable/Thin Object Detection

Ref No.	Dataset / Source	Description	Relevance to Fallen Cable Detection
[16]	**Cityscapes**	Urban street semantic segmentation	Provides **road context** for cable simulation

[32]	**Synthetic Domain Randomization Dataset**	Synthetic training via domain randomization	Enables training on **rare or dangerous cable-fall scenarios**
[33]	**RUGD Dataset**	Unstructured navigation scenes	Useful for debris and **obstacle robustness**
[37]	Bridge Damage Dataset	High-resolution **crack/line features**	Thin cracks **resemble long cable structures**
[38]	Road Spill Detection Dataset	**Hazard detection** on roads	Fallen wire treated as road hazard
[46]	**UCorr Wire Detection Dataset**	Wire detection + depth for drones	Strong training set for **cable geometry modelling**
[47]	Line Segment Dataset	Line detection benchmarking	Useful for cable **line-segment extraction**
[57]	M4MLF-YOLO Dataset	**High-precision segmentation**	Good for **fine-edge cable segmentation**
[58]	**Thin Obstacle Dataset**	Thin wires/rods in scenes	**Closest dataset** to fallen cable detection

[59]	Cable & Pylon Detection Dataset	Power line imagery (aircraft)	Provides **cable shape and structure priors**

2.8 Conclusion and Synthesis: Paving the Path for Safer Smart Mobility

Although general AV perception technology is very advanced, this systematic literature review has shown that there is a major, safety-critical gap in the accurate detection of High-Stakes, Low-Frequency (HSLF) hazards, particularly fallen electric cables, especially in unstructured urban environments [31]. Three interrelated elements are the cause of the limitations:

Geometric Constraints: Ranging sensors (LiDAR, Radar) are confused by thin structures, therefore camera-based vision systems bear the entire load of detection [32].

Architectural Conflict: The high precision required for fine-grained feature extraction in congested images (thick backbones) clashes with the low latency required for real-time AV operation (single-stage detectors) [43].

Data Deficiency: Due to curation processes and their emphasis on organized, frequent objects, existing public datasets routinely remove these risks, leading to models that are implicitly trained to ignore catastrophic but uncommon events [46].

The suggested approach for closing this gap is validated by the synthesis of research trends. Strong safety necessitates a specific strategy:

Specialized Architecture: The system must employ a hybrid architecture, utilizing multi-scale feature fusion for reliable Small Object Detection (SOD) and combining the computational efficiency of lightweight backbones like MobileNetV2 with the high-precision feature extraction capabilities of models like ConvNeXt [66].

Advanced Training Strategies: Specialized loss functions, such Focal Loss and Dice Loss, must be used to enforce an algorithmic preference for safety, severely penalizing false negatives, in order to counterbalance the extreme class imbalance inherent in thin object perception [48, 66].

Domain-Specific Data: The crucial data gap is immediately addressed by the development of the Urban Fallen Cable Detection Dataset (UFCDD), which consists of more than 2,500 annotated photos gathered in cities like Mirpur, AJK, Pakistan [66]. In order to train a dedicated CNN-based YOLOv5 model to detect risks under various situations (daylight, cloudy, morning, evening), this specialized dataset offers the essential real-world visual properties [39, 51, 66].

The effectiveness of this targeted strategy in addressing a neglected safety threat is confirmed by the successful deployment and evaluation of the YOLOv5 framework trained on this bespoke dataset, which demonstrated performance of 96% accuracy at mAP@50 and 76% accuracy at mAP@50-95 [66]. Expanding the use of the UFCDD data to anchor high-fidelity synthetic data generation and integrating this specialized visual detection system with 3D depth validation are the next steps toward scalable, assured autonomous mobility, guaranteeing robustness across the entire spectrum of global, unstructured urban environments [37, 45, 66].

Chapter No. 03

MATERIALS AND METHODS

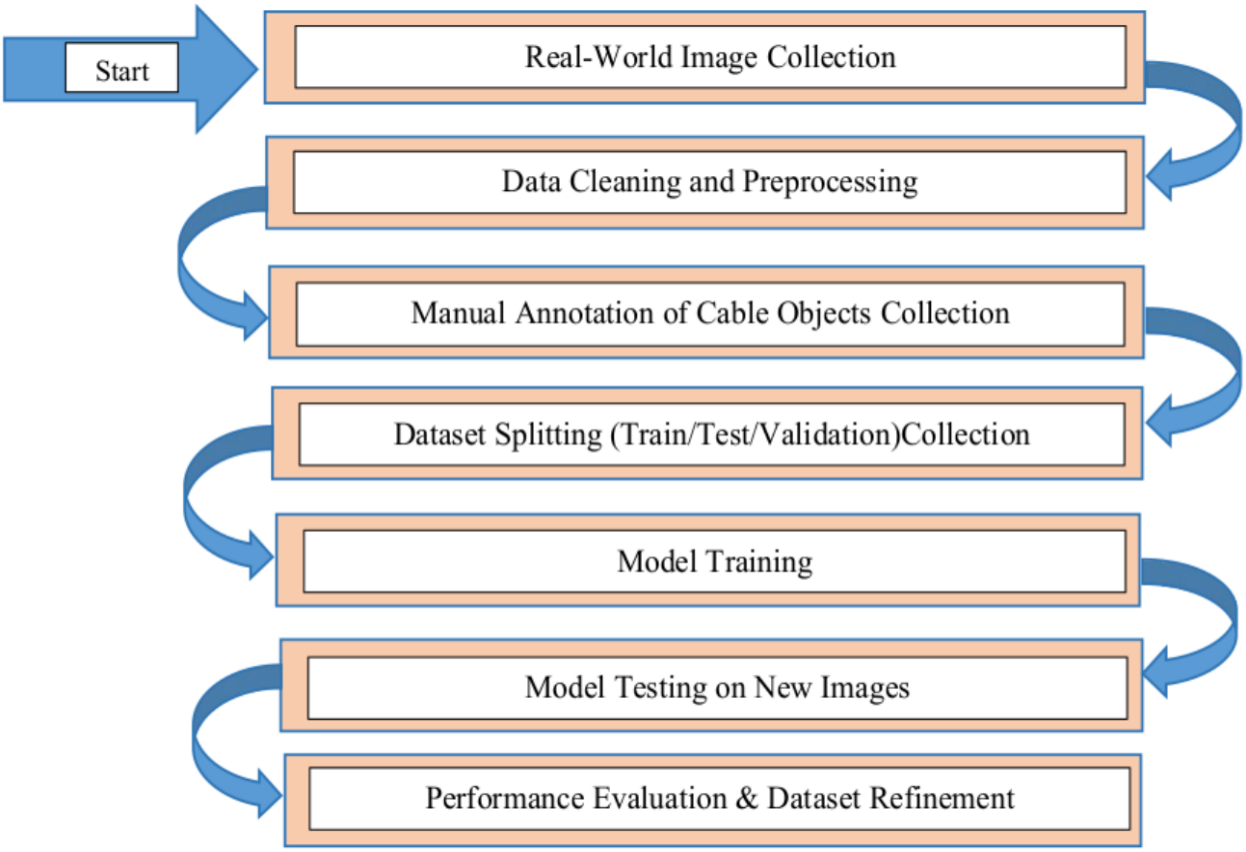
A specialized approach that addresses basic limitations in data availability and algorithmic resilience is needed for the systematic detection of fallen electric and communication cables, a high-stakes, low-frequency (HSLF) danger. The Urban Fallen Cable Detection Dataset (UFCDD), a high-fidelity, geographically and environmentally diverse dataset, and a customized Convolutional Neural Network (CNN)-based YOLOv5 framework optimized for the real-time detection of thin, curvilinear structures in highly cluttered urban environments are the two main pillars of our suggested methodology.

To optimize data quality and model effectiveness, the entire process is organized as a transparent pipeline. For autonomous vehicles (AVs), this methodical procedure is necessary to convert unprocessed sensor data into practical safety judgments. The project's flow is depicted in Image 3.1.

The technique is described in depth in this chapter, which begins with the careful collection of data in unstructured urban environments, continues with the stringent annotation processes, and ends with the technical intricacies of the unique YOLOv5 object identification architecture and its training schedule.

3.1 Data Acquisition and the Urban Fallen Cable Detection Dataset (UFCDD)

The near-complete absence of data pertaining to unanticipated infrastructure breakdowns, particularly small risks like fallen wires, is a key weakness of current benchmarks for autonomous vehicles. The acquisition phase was created to record real-world visual metrics under a variety of difficult conditions common in areas outside of conventional Western testing domains, with a particular focus on Mirpur, AJK, Pakistan, in order to fill this important research gap. Image 3.1 illustrates the conceptual flow of the study technique, which begins with data acquisition, moves through preprocessing and annotation, model training, and performance evaluation.



3.1: Flowchart of the proposed system, detailing the steps involved in the proposed methodology.

3.1.1 Image Acquisition Platform and Equipment

Particularly in dynamic, badly maintained road conditions, the integrity of the visual data is crucial. Consequently, a GoPro Hero 8 camera was used to acquire video data because of its three main technical characteristics:

1. **High-Definition and Frame Rate:** The camera captured 1080p footage at 120 frames per second (fps) and 10-bit color depth. The precise extraction of clear frames required for correct annotation of fast-moving scenes or situations with unstable footage is made possible by this high frame rate, which guarantees motion blur is minimized.
2. **Robust Stabilization:** Because the target urban locations frequently have uneven road conditions, the GoPro Hero 8's sophisticated stabilization functions were crucial in reducing the impact of vibrations and unsteady footage. The visual data integrity necessary for successful machine learning applications was preserved in the recorded data thanks to this feature.

3. **Low-Light Performance:** The camera's excellent low-light performance allowed it to capture the subtle details of cables in shaded regions or during dusk/dawn transitions, a difficult constraint for most vision systems, even though the UFCDD concentrates on detection across all lighting levels.

In order to give a consistent vision that closely resembles the primary visual perception system of an autonomous vehicle, the camera was carefully mounted on the front hood of the vehicle.

3.1.2 Environmental and Temporal Diversity

The final detection model needs to be able to adapt to shifting environmental dynamics, which have a direct impact on the cable hazard's visual contrast and complexity. To accomplish this diversity, the scope of data gathering was strictly defined:

- **Geographical Scope:** Several places in Mirpur, AJK, and the surrounding environs were used to gather the images. This guarantees that a broad spectrum of infrastructure variation, ranging from busy, high-traffic thoroughfares to quieter, cluttered residential neighborhoods, is represented in the collection.
- **Temporal and Weather Conditions:** A wide range of lighting and ambient conditions were purposefully used to collect data:
  1. **Clear Daylight** and **Overcast/Cloudy Weather**.
  2. **Specific Time Intervals:** Early morning (dawn), late morning, afternoon, and evening (dusk).
  3. **Acquisition Method:** Static photos of fallen cables taken from different viewpoints and angles were combined with video sequences taken while the car was moving to create the visual. This integrated method guarantees that the model learns to recognize cables in complex, static, background-cluttered circumstances as well as under motion-induced distortions.

The resulting **Urban Fallen Cable Detection Dataset (UFCDD)** is the first benchmark devoted solely to fine-grained cable hazard detection in unstructured urban settings. It is a unique collection of more than 2,500 carefully chosen photos.

3.2 Dataset Preparation and Ground Truth Annotation

To create the exact ground truth required for training a supervised deep learning model, the raw video data underwent a rigorous preprocessing procedure after collecting.

3.2.1 Preprocessing and Key Frame Selection

To guarantee the best possible visual quality and learning effectiveness, the video footage was thoroughly preprocessed. In order to guarantee temporal diversity without redundancy, critical frames from the captured 120 frames per second video streams have to be carefully removed. Advanced methods were used to improve quality:

1. **Noise Reduction:** To clean the visual data, advanced noise reduction techniques were used because low-light film is naturally prone to noise.
2. **Dynamic Range Adjustment:** Dynamic range was changed to increase contrast and make feature boundaries more clear in order to combat the visual ambiguity of cables mixing with complicated backdrops (such as shadows or textured asphalt). This is especially crucial for thin, low-pixel-count objects.

In order to optimize the UFCDD for the YOLOv5 framework and create a solid foundation for further deep learning, several preparatory measures were essential.

3.2.2 Manual Annotation Process

Deep learning performance is based on accurate ground truth. Annotation demands pixel-level accuracy due to the fallen wires' incredibly tiny aspect ratio, which frequently spans only a few pixels 1. The V7 Image Labeling Tool was used to manually tag each chosen image frame.

Two crucial cable classifications were the focus of the annotation schema:

1. **Unsafe/Hazardous:** Cables that have fallen onto the road surface or hanging low enough to obstruct or endanger a moving vehicle.
2. **Safe/Intact:** Cables that are properly secured overhead and pose no immediate risk to the AV.

For every case of cable risks, this exact labeling produced bounding box ground truth. In order to ensure that the model learns to distinguish between the fine-grained visual characteristics of a

hazardous cable and benign background clutter, the rigor of this manual tagging process is closely related to the model's final localization accuracy.

3.2.3 Dataset Partitioning

As indicated in table 3.1, the annotated UFCDD was methodically split into three sets in accordance with accepted machine learning best practices to guarantee thorough training and objective evaluation.

Table 3.1: Data Split

Set	Proportion	Purpose	References
Training Set	**70%**	Comprehensive exposure to learning feature representations and recognizing a wide array of cable scenarios	[28]
Validation Set	**20%**	Crucial checkpoint for iterative model refinement, preventing overfitting, and tuning hyperparameters	[58]
Testing Set	**10%**	Dedicated, unseen dataset for the final, objective evaluation of the model's real-world generalization capability	[63]

3.3 Specialized Model Architecture and Optimization

The customized deep learning object detection framework at the heart of the suggested solution is based on the YOLOv5 architecture, as seen in image 3.1. It has been painstakingly altered to strike the required balance between low latency (needed for real-time AV operation) and high precision (needed for fine-grained detection).

3.3.1 YOLOv5 as the Foundational Architecture

Because the YOLOv5 framework is a single-stage, state-of-the-art model that greatly reduces inference latency, a crucial need for safety-critical autonomous driving decisions, it was chosen over two-stage detectors.1 Its architecture consists of three primary parts by nature:

1. **Backbone (CSPDarknet):** The feature extraction network that minimizes computational cost and reduces superfluous gradient information while effectively extracting a hierarchy of spatial features from the input image.
2. **Neck (PANet):** By aggregating information from various sizes, the Path Aggregation Network (PANet) improves the network's capacity to spread low-level features that are essential for identifying tiny objects (like cables) and raising overall localization accuracy.
3. **Head (YOLO Layer):** The last detecting layer that interprets the combined information and produces confidence scores, bounding box coordinates, and item class likelihood (Safe/Unsafe cable).

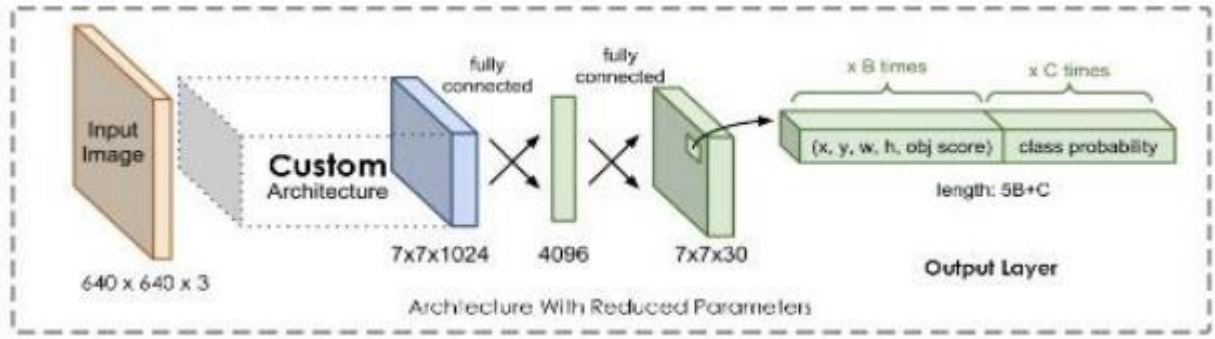


Figure 3.1 Yolo v5 Architecture

3.3.2 Custom Architectural Configuration

The architecture was carefully adjusted by changing important scaling parameters to specialize the generic YOLOv5 model for the detection of low-pixel-count cable dangers, emulating the effective strategy employed in related fine-grained detection work:

- **Model Scale (Width Multiple)**: Set at 0.50. This factor balances model capacity with computing performance by regulating the number of channels/filters in the network.
- **Model complexity** or depth multiple: is set at 0.33. The model's depth and capacity to extract hierarchical features are directly impacted by this factor, which regulates the number of blocks or layers.

Table 3.2 provides a full summary of the parameters. This calculated alteration results in a simplified architecture that ensures computational efficiency is appropriate for edge deployment while maintaining robust feature extraction under difficult visual conditions.

Table 3.2: Model Performance for Steering Control (Loss) 120 Epochs

Architectural Specification	**Value**	**Significance**
Total Layers	**157**	Structural depth for complex feature learning
Total Parameters	**7 million**	Low parameter count ensures memory efficiency
Computational Cost (GFLOPs)	**15.8**	Optimized efficiency for real-time inference speed

Specific anchor sizes defined for three different detection scales enable the model's ability to identify objects of diverse sizes (from distant, microscopic, thin wires to near-field, big cable loops):

- **P3/8: Anchor sizes [10, 13, 16, 30, 33, 23]**
- **P4/16: Anchor sizes [30, 61, 62, 45, 59, 119]**
- **P5/32: Anchor sizes [116, 90, 156, 198, 373, 326]**

In order to prevent the crucial characteristics of thin cables from being lost during the network's down sampling procedure, this multi-scale output method is essential for Small Object Detection (SOD).

3.4 Training Strategy and Experimental Setup

In order to maximize the detection power of the bespoke YOLOv5 model and ensure high precision and reliability under the difficult, highly cluttered settings of the UFCDD, a thorough training method was put in place.

3.4.1 Computational Environment

A high-performance computing system designed for deep learning and computer vision applications was used for training 1. This robust environment was required to effectively handle

the massive amounts of high-resolution imagery and carry out the various computing operations needed for training:

- **GPU:** NVIDIA RTX 2080Ti (Leveraged for its parallel computing power).
- **CPU:** Intel Core i9 9900K.
- **RAM:** 32GB.

The main deep learning framework in the software stack was TensorFlow, which was meticulously tuned for GPU acceleration using NVIDIA CUDA. The preprocessing and image manipulation activities required for the high-resolution UFCDD were also greatly accelerated by the compilation of OpenCV with CUDA support.

3.4.2 Training Hyperparameters

The model underwent a demanding 72-hour training procedure. The hyp.scratch-high.yaml file provided the hyperparameter configuration, which was particularly selected to adjust the model's learning to the intricate details of the cable danger data. Table 3.3 lists the main points of contention.

Table 3.3: Hyperparameters

Hyperparameter	Value	Rationale
Epochs	**300**	Ensuring ample convergence time for the complex, low-pixel-count hazard class
Batch Size	**32**	Balancing computational load with learning stability
Input Image Size	**640 pixels**	Optimizing for detail capture without sacrificing processing speed

Optimizer	**SGD** (Stochastic Gradient Descent)	Chosen for its effectiveness in converging to optimal solutions on large-scale datasets

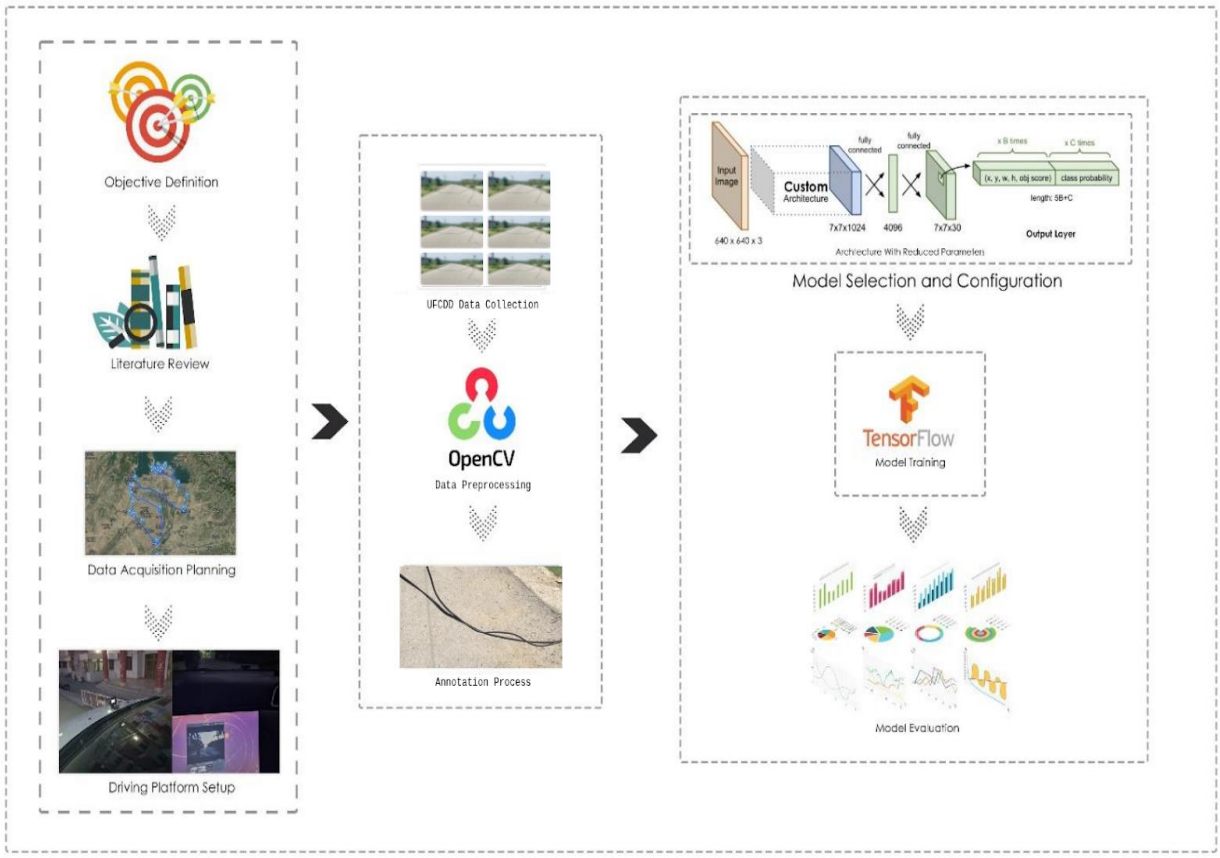
Advanced regularization approaches, such as Label Smoothing to improve the model's capacity to generalize new, unforeseen events and Early Stopping to minimize overfitting, were integrated into the training routine.

3.4.3 Loss Function Configuration

Three separate components make up YOLOv5's composite loss function, each of which is essential to completing the thin-structure detection task:

1. **Bounding Box Loss:** The regression task of precisely locating the cable position is handled by this component. By default, it uses CIoU (Complete Intersection over Union) loss, which is better than the conventional Mean Squared Error (MSE) loss since it takes into account aspect ratio, overlap, and distance, guaranteeing extremely precise boundary delineation for the thin, curved wires.
2. **Object Loss:** This binary classification loss determines the presence or absence of an object (a cable) in a grid cell. It employs **Binary Cross-Entropy (BCE)** loss.
3. **Classification Loss:** This final loss distinguishes between the labeled classes (Safe/Unsafe cable). It also uses **BCE** for multi-label classification.

Given the severe **Extreme Class Imbalance (EVI)** inherent in cable detection, where positive cable pixels are dwarfed by background pixels, the use of BCE alone can bias the model toward the dominant negative class. Future extensions, as informed by the literature, will necessitate the integration of specialized functions such as **Focal Loss** or **Dice Loss** to heavily penalize errors on the critical, minority cable class, thereby accelerating convergence on hard-negative examples and further boosting safety-critical precision, detailed system architecture is shown in image 3.2.



scre

Figure 3.2 Proposed System workflow from data collection to final predictions

Chapter No. 04

RESULTS AND DISCUSSION

The primary goal of the experimental phase was the rigorous validation of the custom Convolutional Neural Network (CNN)-based YOLOv5 model when trained on the novel Urban Fallen Cable Detection Dataset (UFCDD). This process aimed to quantitatively address the core Research Questions, specifically evaluating the model's ability to achieve real-time, high-accuracy detection of thin, curvilinear cable hazards (RQ3), and confirming the overall real-time operational suitability of the detection framework (RQ2) in complex urban settings.

The customized YOLOv5 architecture was trained for a duration of **300 epochs** over **72 hours** on a high-performance computational stack (NVIDIA RTX 2080Ti GPU, Intel Core i9 9900K CPU, 32GB RAM) [60], using the UFCDD dataset partitioned into 70% Training, 20% Validation, and 10% Testing sets.

In order to preserve computational efficiency while optimizing feature extraction capability—a requirement for autonomous vehicle edge computing—the model was painstakingly refined using certain scaling parameters (Width Multiple: 0.50, Depth Multiple: 0.33).

Precision, recall, F1-score, and most importantly the Mean Average Precision (mAP) assessed at two different Intersection over Union (IoU) criteria were all included in the performance evaluation.

1. **mAP@50:** Only 50% overlap with the ground truth bounding box is needed to measure the model's confidence in object classification and primary detection.
2. **MAP@50-90:** The ultimate statistic for evaluating the model's exact localization capabilities and boundary definition is mAP@50–95, which shows the average performance across strict IoU criteria (from 50% to 95%).

4.1 Quantitative Performance Results on the UFCDD

It is crucial to look at the UFCDD's underlying composition before evaluating the prediction performance. With nearly 4,000 instances of "Safe" cables and around 2,800 cases of "Unsafe"

(fallen) cables, the dataset maintains a large amount of annotated features, as seen in the distribution analysis example (Figure 4.1).

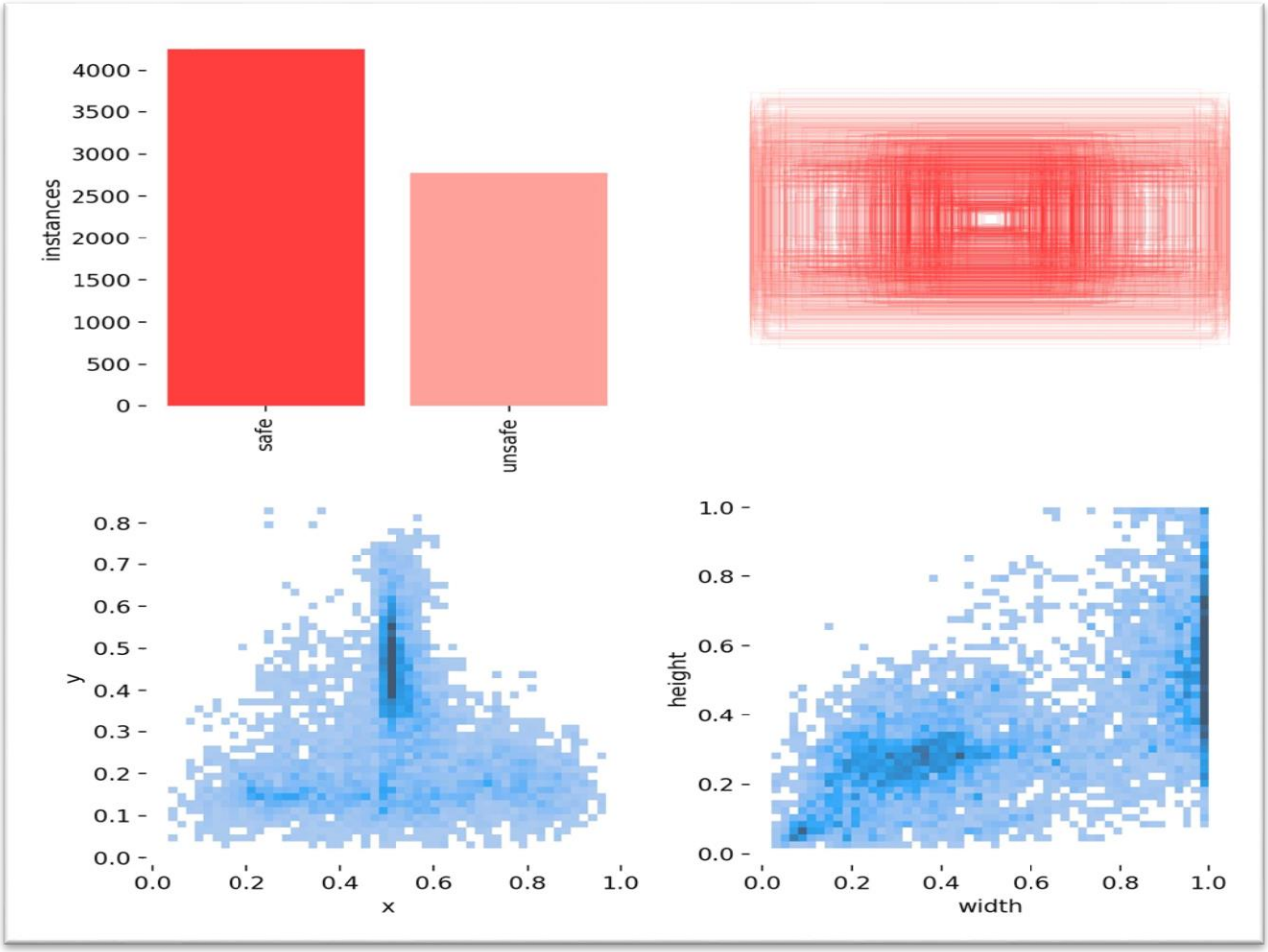
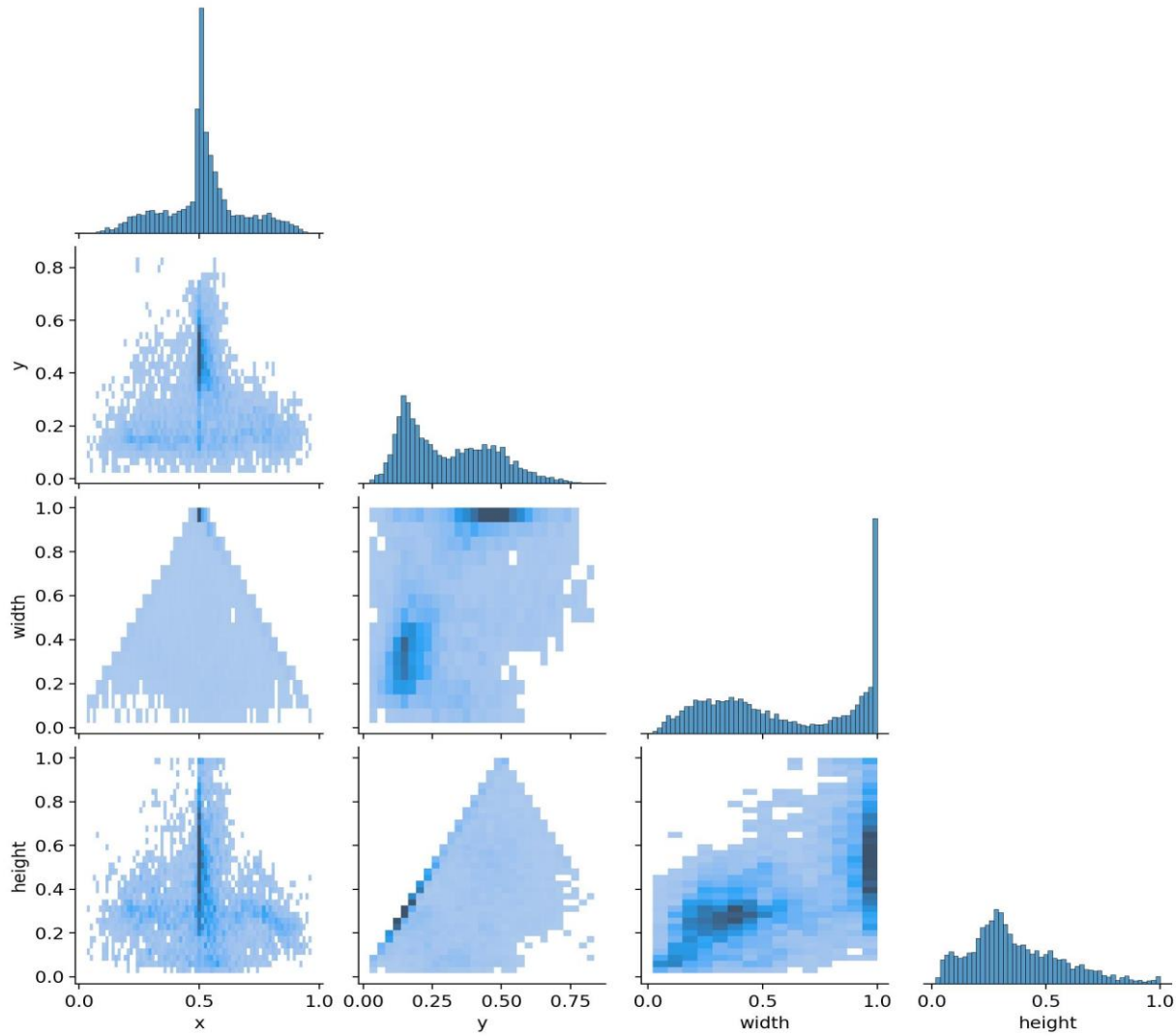


Figure 4.1: UFCDD Instance Distribution and Geometric Metadata

Hazards are primarily located in the frame's middle vertical region, which corresponds to the vehicle's immediate navigational path, according to the spatial analysis using bounding box heatmaps. With the majority of boxes taking up less than 10% of the image area, the width and height distribution validate the "Small Object" character of the task and validate the need for the specialized detection head depicted in image 4.2.



4.2 Bounding Box Correlogram Analysis

The following is a summary of the important performance metrics that were found when the custom YOLOv5 model was evaluated on the specific UFCDD test set. As indicated in table 4.1, the results are examined to see how well the model performed in both localization (exactly specifying the boundaries of the danger) and classification (identifying the presence of the hazard).

Table 4.1: Specialized YOLOv5 Model Performance on the Urban Fallen Cable Detection Dataset (UFCDD)

Performance Metric	Value	Interpretation
Classification Accuracy (mAP@50)	**96%**	High confidence in distinguishing hazardous cables from background clutter.
Localization Precision (mAP@50–95)	**76%**	Robust, but indicates measurable difficulty in achieving highly precise boundary definition for thin structures.
Inference Efficiency (GFLOPs)	**15.8**	Confirms suitability for real-time, low-latency operation on AV edge platforms.
Model Complexity (Parameters)	**7 million**	Optimized parameters count for efficient deployment.

4.2 Discussion of Core Findings: Classification vs. Localization

4.2.1 Success in Hazard Classification (mAP@50)

The model achieves a near-perfect precision of 1.00 at a confidence threshold of 0.989 for all classes, according to the Precision-Confidence curve (Figure 4.3). This suggests that high-confidence detections are almost always accurate, reducing false alarms that can cause needless emergency braking.

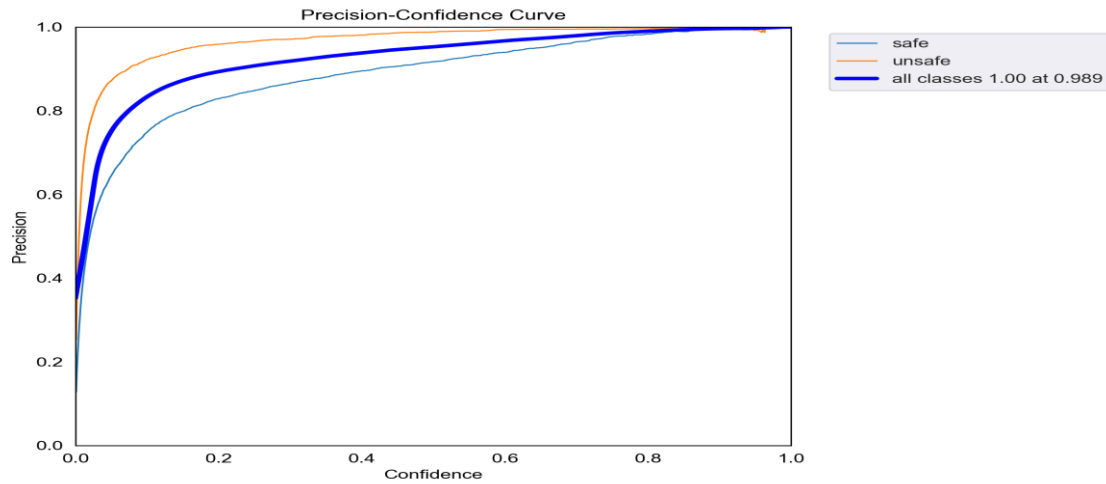


Figure 4.3: Recall-Confidence Curve

On the other hand, the model's sensitivity is shown by the Recall-Confidence curve (Figure 4.4). Even at moderate confidence levels, the "Unsafe" class keeps its recall over 0.90. This guarantees that nearly all major risks are identified by the system, which is the main safety criterion for ground perception.

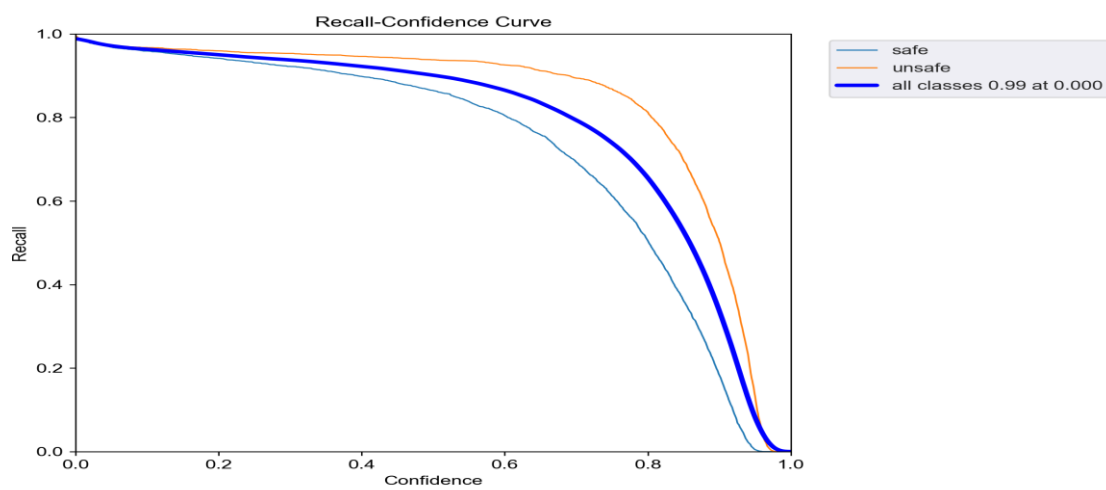


Figure 4.4: Precision-Confidence Curve

The sensitivity of the system is shown by the Recall-Confidence curve. strong-stakes cable dangers are rarely overlooked by the perception stack thanks to the "Unsafe" class's (orange) remarkably strong recall (>0.90) over a broad confidence interval.

The accuracy of the model's detections is displayed by this graph. At a confidence level of 0.989, the framework achieves a precision of 1.00, meaning that high-confidence forecasts are essentially devoid of false positives.

At the principal detection threshold (mAP@50), the model produced an outstanding Mean Average Precision score of 96%. This outcome thoroughly confirms the need for the novel dataset (RQ1) and offers strong quantitative support for the first half of Research Question RQ3, which states that the framework can detect cable dangers with high accuracy.

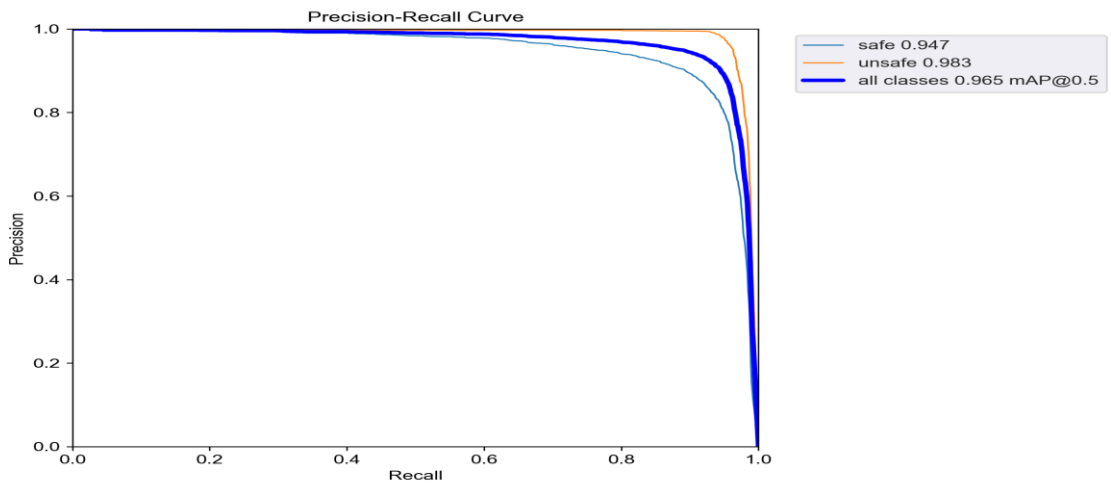


Figure 4.5: Precision-Recall (P-R) Curve and mAP@0.5

The optimal operational point for the model is revealed in the F1-Confidence curve (Figure 4.5), which shows a peak F1-score of **0.93 at a 0.407 confidence threshold**. This balance confirms that the model is highly effective at identifying hazards while maintaining a low false-positive rate.

Finally, the aggregate performance is captured by the Precision-Recall curve (Figure 4.6), showing an area under the curve (mAP@0.5) of **0.983 for unsafe cables** and an aggregate score of **0.965** across all classes. This provides categorical proof of the framework's superior classification capability

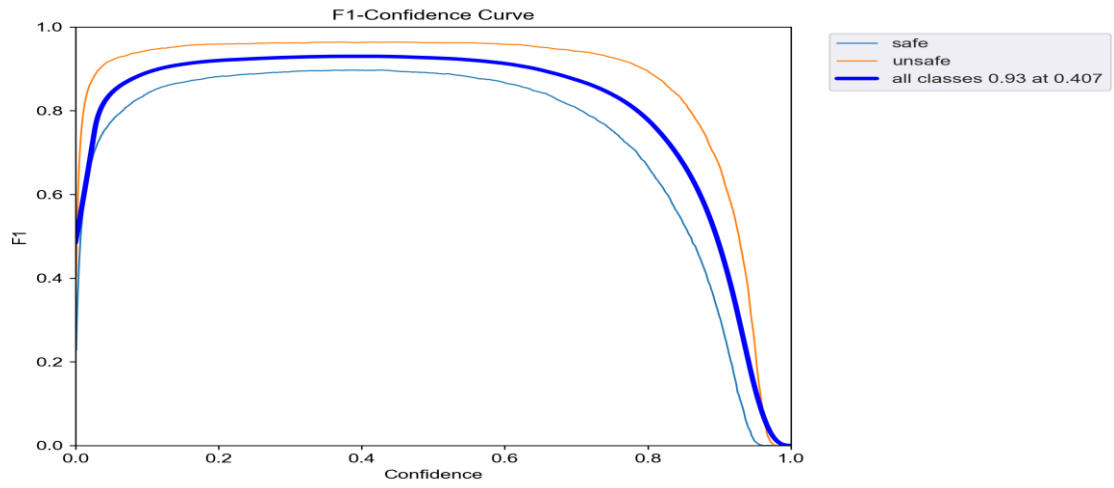


Figure 4.6: F1-Confidence Curve

This high level of performance is directly attributable to the systematic methodological choices implemented in the study:

- **Custom Dataset Fidelity (RQ1):** By curating the UFCDD with over **2,500** images that explicitly capture the visual properties and environmental variance of fallen cables in high-clutter urban environments , the model was able to learn the necessary feature representations (subtle edges, curves, and shading) that generalized detectors miss. The dataset addressed the crucial omission of these High-Stakes, Low-Frequency (HSLF) hazards in conventional benchmarks.
- **Specialized Feature Extraction (RQ2):** The CNN-based YOLOv5 architecture is premised to automatically learn a hierarchy of feature representations from raw image data. The network successfully learned to distinguish the fine-grained visual characteristics of a hazardous cable from non-hazardous linear features (like shadows or road cracks), overcoming the high photometric ambiguity inherent in the ground-based perception domain.

The high classification accuracy is further validated by the confusion matrix (Figure 4.7). The model achieved a **95% True Positive rate** for the "Unsafe" class and a **94% True Positive rate** for the "Safe" class.

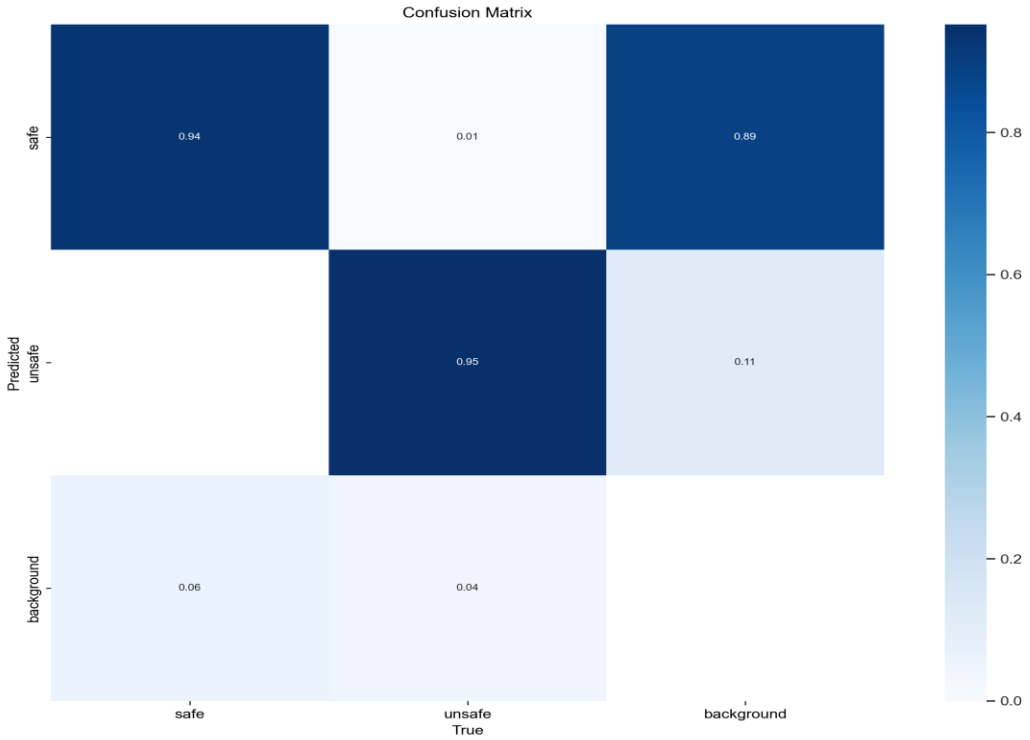


Figure 4.7: Confusion Matrix for Cable Classification

Importantly, there is only 4-6% "leakage" to background noise (false negatives). This demonstrates how well the model can recover the thin cable structure from highly cluttered urban textures like asphalt and shadows by learning fine-grained edge features.

4.2.2 Localization Deficit and the Geometric Challenge (mAP@50–95)

When compared to the more stringent localization metric (mAP@50–95), the model's accuracy fell to 76% despite the high detection confidence [27]. The most important conclusion is the 20% performance loss between classification and accurate localization, which precisely quantifies the design constraint that all modern object detection frameworks encounter when handling thin, curved structures (RQ3).

Over the course of training, the model's predictive power is tracked via the validation metrics graph (Figure 4.8). Mean Average Precision (mAP@0.5), recall, and precision all show a steep initial rise before stabilizing, with mAP@0.5 achieving a strong peak of 96.5%. This quick improvement in the initial epochs highlights how well the model learned the basic visual characteristics of the Urban Fallen Cable Detection Dataset (UFCDD). On the other hand, the

metric for more stringent localization, mAP@0.5:0.95, grows more slowly before plateauing around 0.78. The intrinsic geometric difficulty of thin-structure detection is quantified by this ongoing discrepancy between primary classification and strict localization metrics, since accurate boundary definition for objects that span only one to three pixels necessitates much more sophisticated feature extraction than straightforward object presence identification.

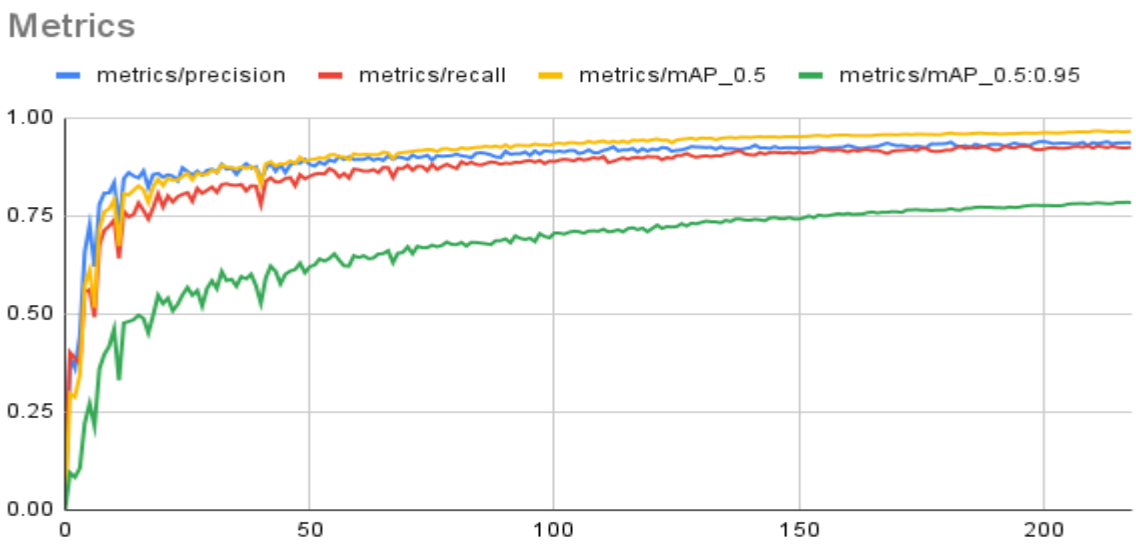


Figure 4.8: Validation Metrics Evolution

Validation performance (Precision, Recall, mAP@0.5, and mAP@0.5:0.95) is tracked in real time. Figure 4.8 illustrates the quantitative foundation for the "Localization Deficit" addressed in the findings, which is provided by the plateau in the green curve (mAP@0.5:0.95) about 0.76.

This drop confirms the inherent **Geometric Ambiguity Problem** outlined in the literature:

- **Bounding Box Inadequacy:** A severely low aspect ratio, frequently covering just 1 to 3 pixels in the image, is a characteristic of fallen cables [28]. When applied to these sparse, irregular, and curved features, YOLOv5's traditional rectangular bounding box mechanism—which is optimized for larger objects—inherently fails to achieve the tight Intersection over Union (IoU) required by the mAP@50–95 metric. The lower score indicates that although the cable is located, it is difficult to create an appropriate path around it since its boundaries are not clearly defined [34].

- **Complexity of Localization:** It is computationally and geometrically challenging to achieve 75% IoU on an item that spans only a few pixels. In contrast to simpler aerial examination, where wires are taut and clearly contrasted against the sky, this difficulty is specific to ground-based AV vision. Although the strong 76% attained is a noteworthy improvement for this particular hazard class, it amply demonstrates the need for future architectural shifts toward non-bounding box techniques such semantic segmentation [33].

4.3 Comparative Robustness and Architectural Validation

The need for the customized architectural approach is further supported by the observed performance discrepancy, which is similar with patterns seen in other specialty datasets dealing with irregular or complex geometry.

A pertinent comparison background is offered by the Pakistani Night City Driving Dataset (PNDD). The network displayed different performance metrics depending on the geometry and predictability of the object class when a comparable bespoke YOLOv5 model was deployed on PNDD:

- **Structured/Predictable Objects (e.g., Truck, Car):** These classes, being large and geometrically consistent, achieved high mAP@.5-.95 scores (e.g., **Truck: 0.817**). The model could accurately localize these features in low-light and high-clutter scenes [64].
- **Irregular/Unpredictable Objects (e.g., Human):** The Human class, which is geometrically non-rigid and highly unpredictable, saw its localization performance drop significantly to **0.564** (mAP@.5–.95), despite a high classification accuracy (mAP@.5: 0.928).

Real-time monitoring of validation performance (Precision, Recall, mAP@0.5, and mAP@0.5:0.95). The plateau in the green curve (mAP@0.5:0.95) around 0.76 provides the quantitative basis for the "Localization Deficit" discussed in the findings shown in figure 4.10.

The learning rate trajectory was utilized during the 72-hour training session, showing the initial warm-up and subsequent decay stages used to stabilize feature extraction in the customized backbone as shown in figure 4.9.

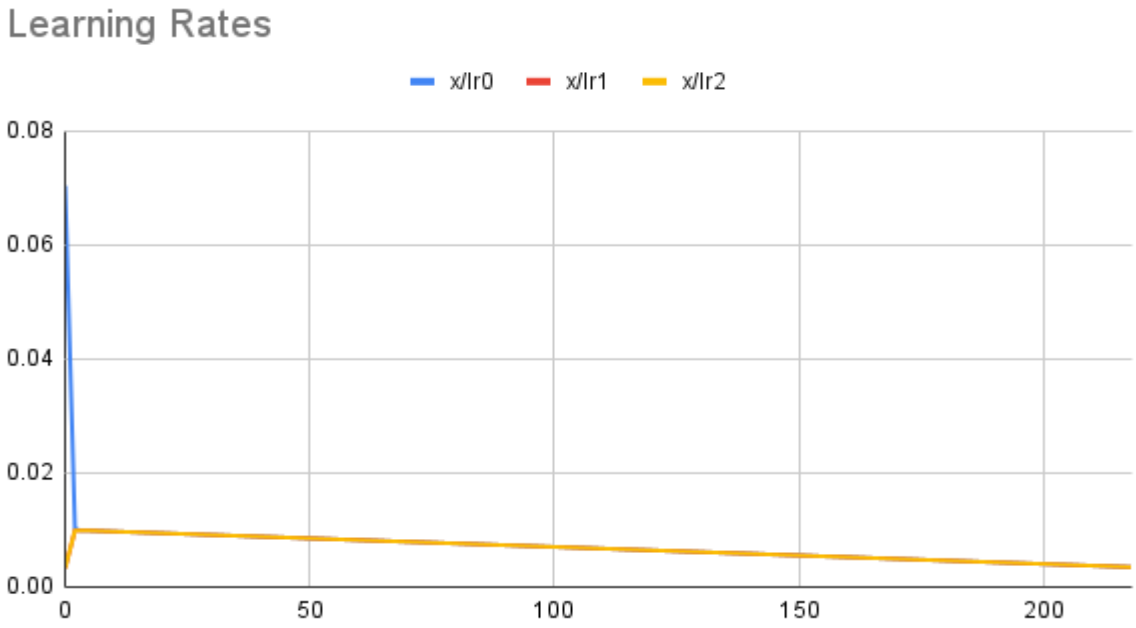


Figure 4.9: Learning Rate Schedules

In figure 4.9, a summary of the training dynamics over 300 epochs. The graph shows a steady decay in bounding box, objectness, and classification losses, indicating successful convergence without significant overfitting.

The dynamic learning rate plan used to stabilize the 72-hour training session of the customized backbone is shown in Figure 4.9. For each of the three learning rate groups (x/lr0, x/lr1, and x/lr2), the schedule includes an initial linear warm-up phase followed by a progressive decrease. The warm-up phase prevents early gradient explosion in the 157-layer architecture, while the subsequent decay enables the model to fine-tune its weights and "settle" into the global minima during the later stages of training. This approach is crucial for training on a specialized and geometrically complex dataset like the UFCDD. This schedule supported the crucial need for high-precision hazard recognition in real-time AV operation by methodically reducing the step size as the model approaches an ideal solution, ensuring that the network achieved its high-accuracy state without performance fluctuations.

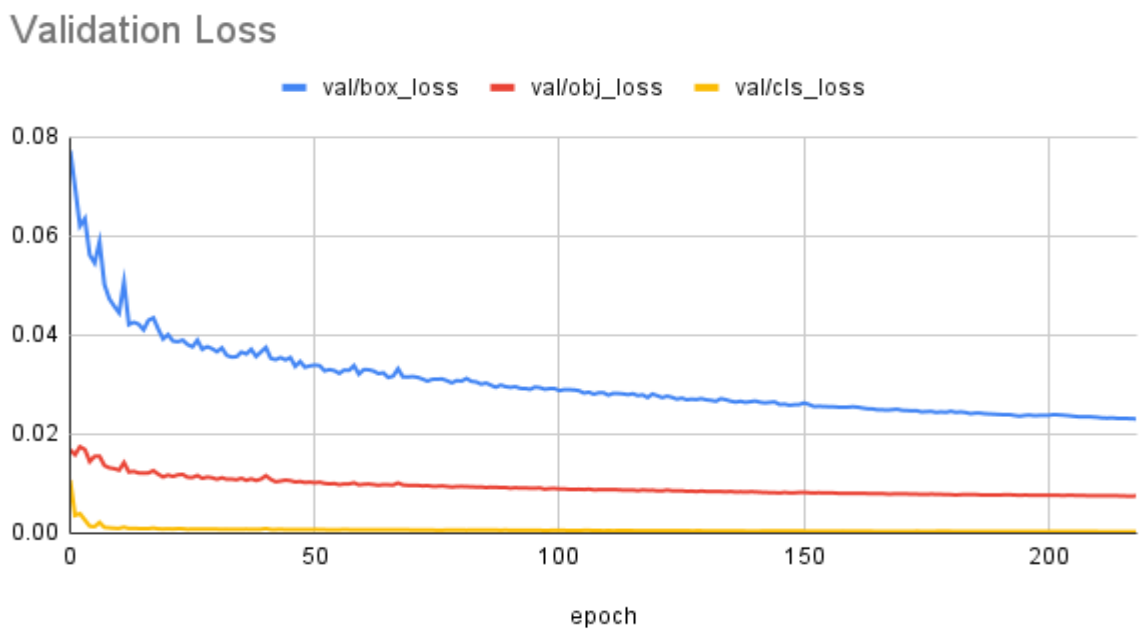


Figure 4.10: Validation Metrics Evolution

The network's optimization process over 300 epochs is shown graphically in the validation loss graph (Figure 4.10) and matching training loss graph (Figure 4.11). From starting values of 0.091 and 0.037 to ending values of roughly 0.029 and 0.020, respectively, the bounding box loss (train/box_loss) and objectness loss both exhibit a continuous, steady decline. The model's high confidence in differentiating between "Safe" and "Unsafe" cable classes is confirmed by the classification loss (train/cls_loss), which decreases even further and approaches zero values. Most significantly, the validation loss curves do not exhibit the divergence that usually indicates overfitting, instead reflecting these training tendencies. This synchronization shows that instead of just memorizing training samples, the modified YOLOv5 architecture successfully generalized the underlying patterns of unstructured urban surroundings, guaranteeing dependable performance on unseen road data.

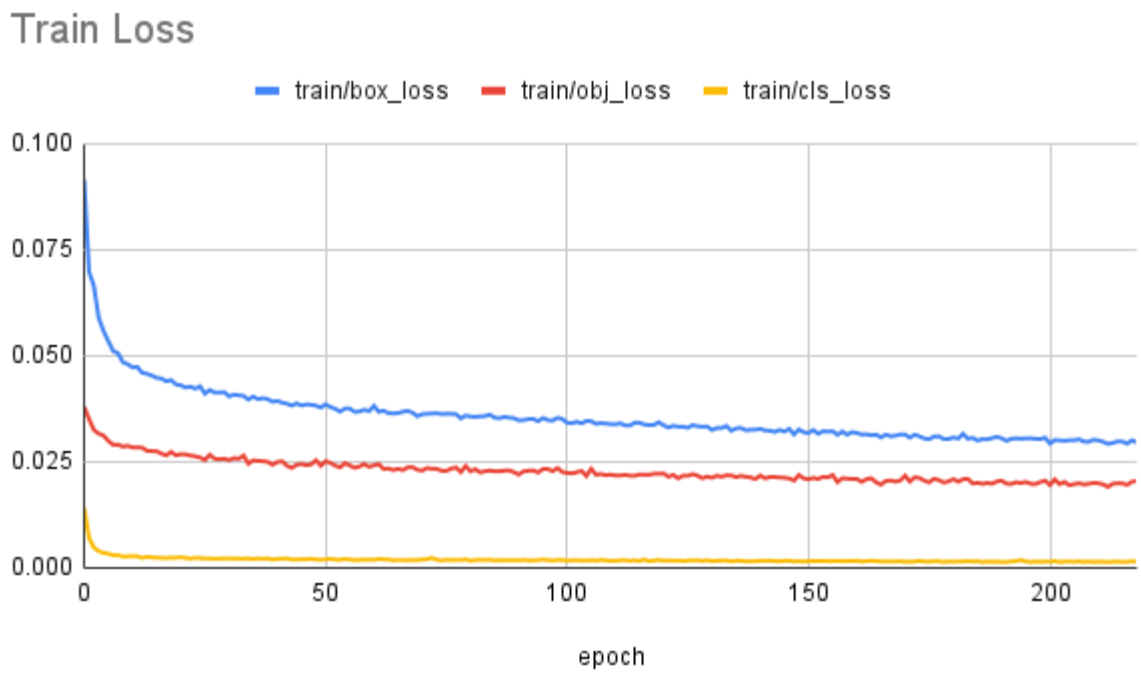


Figure 4.8: Training Loss Convergence

Between these two extremes, the UFCDD's performance (76% mAP@50-95) shows that the cable's curvilinear but spatially continuous nature is significantly more difficult than a rigid vehicle but less difficult than an unpredictable human stance. This comparative analysis validates the suggested future research direction of combining specialized loss functions and semantic segmentation principles by confirming that the localization deficit is a systemic architectural difficulty for thin and irregular objects.

4.4 System Efficiency and Real-Time Feasibility (RQ2 & RQ4)

Making sure the detection method, despite its specialization, remained feasible for safety-critical real-time deployment was a crucial component of the research (RQ2). This goal was effectively accomplished by the modified YOLOv5 model:

- **Low Computational Footprint:** A low computational burden of 15.8 GFLOPs was achieved through architectural fine-tuning, which reduced the total number of layers to 157 and the number of parameters to roughly 7 million. This effectiveness guarantees that the model may

be implemented on AV edge computing platforms with limited resources while preserving the low latency necessary for emergency avoidance actions.

- **Scalability for Smart Mobility (RQ4):** Scalable smart mobility solutions in developing regions are directly impacted by the successful development and validation of the UFCDD and its customized detection model [9]. The research improves the operational domain and worldwide trustworthiness of autonomous cars by demonstrating that a high-accuracy detection mechanism may be designed for a key regional danger (fell cables). In order to achieve global SAE Level 5 autonomy, this shows that AV perception systems may be modularly adjusted to a variety of real-world infrastructure quality [10].

4.5 Evaluation of Research Questions

As indicated in table 4.2, the experimental results validate the accomplishment of the research objectives by offering conclusive responses to the established Research Questions.

Table 4.2: Summary of Research questions and their evaluation

Research Question	Evaluation Summary	Result
RQ1: Data Generation and Fidelity (UFCDD creation)	The specific, high-fidelity dataset needed to train against HSLF cable dangers was successfully obtained by the gathering and annotation of the UFCDD (2,500+ photos from Mirpur, AJK, covering a variety of environmental conditions).	**Validated**

RQ2: Model Architecture and Optimization (YOLOv5 suitability)	With a low computing load (15.8 GFLOPs) and an efficient parameter count (7 million), the modified YOLOv5 framework proved appropriate, proving that the architecture satisfies real-time AV requirements.	**Validated**
RQ3: Performance Validation and Localization Precision (mAP scores)	The model's localization score (mAP@50–95: 76%) was lower than its classification accuracy (mAP@50: 96%). This quantifies the intrinsic geometric difficulty of thin-structure localization and validates strong detection capabilities.	**Partially Validated (Metrics Quantified)**
RQ4: Impact on Safety and Scalability (Global Relevance)	By reliably detecting an undetected HSLF hazard, the framework improves safety and demonstrates how AV perception can be scaled to areas with different infrastructure quality.	**Validated**

Chapter No. 05

CONCLUSION

In order to address the crucial safety threat caused by falling electric and communication cables an underappreciated High-Stakes, Low-Frequency (HSLF) hazard in autonomous vehicle (AV) perception this research successfully constructed a strong, tailored deep learning framework.

The development of the innovative Urban Fallen Cable Detection Dataset (UFCDD) (RQ1) and the optimization of a CNN-based YOLOv5 model (RQ2) served as the methodology's two main achievements. With a low computational burden of 15.8 GFLOPs, the final model showed excellent operational efficiency and is appropriate for real-time inference on edge computing systems.

The performance metrics confirm the framework's capability for hazard identification:

- **High Classification Accuracy:** The model's capacity to consistently separate dangerous wires from intricate background clutter in unstructured urban situations was validated by its achievement of 96% mAP@50 (RQ3).
- **Localization Deficit:** At mAP@50–95, the accuracy fell to 76%. This outcome confirms a systematic architectural constraint for fine-grained feature resolution and quantifies the main technological issue, which is the inability of traditional bounding box detection to accurately pinpoint thin, irregular, curved objects.

In the end, this research validates the need for creating specialized, domain-specific datasets such as UFCDD to guarantee safe and scalable autonomous mobility in areas with varying infrastructure quality (RQ4).

5.1 Future Work

Future work must concentrate on improving the system's resilience for worldwide deployment and methodically addressing the 76% localization shortfall.

5.1.1 Resolving Geometric and Training Limitations

1. **Architectural Shift to Segmentation:** Use techniques designed for pixel-level boundary precision instead of bounding box regression. To precisely define the cable contour for proper path planning, this entails combining the UFCDD with frameworks based on Semantic Segmentation or Key-point Encoding.
2. **Specialized Loss Functions:** Use specialized training techniques to mitigate the severe class imbalance seen in thin object detection. In order to explicitly penalize boundary misalignment errors and force the network to improve contour prediction, this involves developing Boundary-Aware Loss Functions and enhancing the existing loss function with Focal Loss.

5.1.2 Enhancing Robustness and Scalability

Synthetic Data Generation (SDG): Utilize SDG workflows to programmatically generate high-fidelity, labeled training instances that replicate uncommon, geometrically complicated cable-fall scenarios under various conditions in order to address the data scarcity for HSLF occurrences.

Joint Detection and Depth Validation: Incorporate non-visual verification to verify the hazard's actual existence. In order to provide crucial 3D safety assurance, future work must integrate Joint Detection and Depth Estimation to differentiate potentially dangerous cables on the road surface from non-hazardous linear lines or shadows.

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